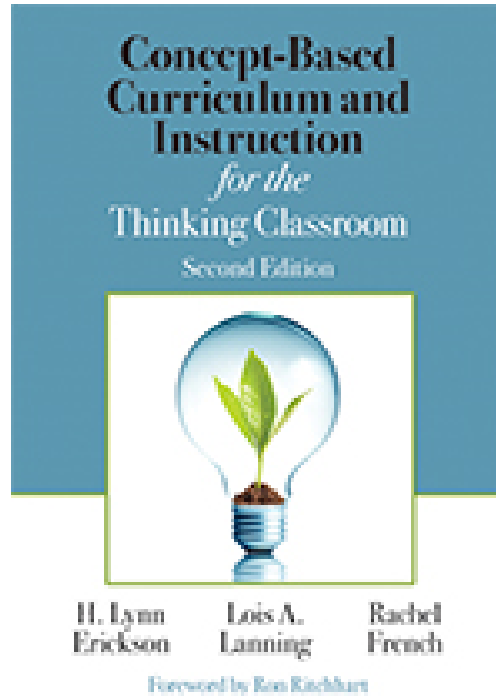


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Foreword

It is the beginning of the school year, which means it is also the time for opening talks to teachers—talks aimed to engage and inspire the troops to participate fully in the important task of teaching that awaits. Having participated in many of these events, both as a teacher myself and as a speaker, I've noticed that opening talks tend to fall into three main categories based on their intent: motivation, inspiration, or commitment. Of course, a good speech will seek to operate on all three fronts simultaneously. However, this isn't always possible as the context of the speech, as well as both the advance and future planning around it, establishes the parameters of any speech's potential. An examination of the three types of speeches will reveal these limiting factors.

First off, we have the motivation speech. At these events, and they are more events than professional learning, teachers will be entertained by a speaker brought in ostensibly to make them laugh. These gatherings often resemble a pep rally, complete with cheering, aimed at pumping energy into what must be perceived as teachers in need of coaxing back to work after their summer vacation. These sessions may also be big on community building, bonding experiences, and motivation. Since the speakers' credentials are often based on their engaging personas more than their worthwhile ideas, there is little chance that these events will make much of a difference to teaching and learning in the long run. They produce a lot of hype but not much substance. Furthermore, since the event has been planned as a one-time kick-off for the school year, there is frequently no thought to leveraging it beyond the moment.

Next, we have the inspiration speech. Here, an outside speaker, often an author, will seek to inspire with new ideas that have taken hold in the minds of the school leadership but that in fact have little chance of catching fire beyond the short-lived moment of introduction. The ideas presented, as good as they may be, become fruit for the picking by those few souls willing to do so. The hope is that some will be as inspired as the school leader is and somehow movement, change, and progress will happen. However, the school leadership has generally not made plans for a robust harvesting of the fruit. What is more, the leadership may not have even considered tilling the field and preparing it to make sure seeds can take hold and grow throughout the year.

In a few venues, the opening talk will have quite a different character altogether. It will be about commitment. In these settings, the beginning-of-the-school-year talk will be an opportunity to demonstrate a renewed dedication to a set of ongoing goals to which the school is committed for the long term. For such a speech, the speaker has been carefully chosen because of his or her connection to the school's master plan. What is more, in these settings, teaching is viewed as a complex process that one engages in collectively rather

than just individually. Teaching is seen as a craft that blends artistry and science, as something that one never fully masters. In these school settings, there is an expectation that teachers have a growth mindset toward their craft, that is, that they view teaching as something that one gets better at and improves on constantly as a result of the challenges one takes on, the risks one takes, and the efforts one puts forth. It is within this context that the speaker is likely to accomplish all three goals—motivation, inspiration, and commitment. Not surprisingly, it is also the commitment speech that is most likely to be remembered and to have a lasting impact on the teaching and learning that takes place in a school.

In reading *Concept-Based Curriculum and Instruction for the Thinking Classroom* by H. Lynn Erickson, Lois A. Lanning, and Rachel French, I am reminded of these three types of back-to-school speeches precisely because this is a book that simultaneously inspires, motivates, and provokes commitment to the craft of teaching. It does this by focusing on big ideas worthy of attention and by honoring teaching and learning, treating them as the astoundingly complex endeavor that they are.

The complexity of teaching makes it an endeavor worthy of our commitment, energy, and passion. While teaching is engaging, fascinating, and often joyous, it is nonetheless complex, and rather than denying or running away from that complexity, we must embrace it. Teaching is complex in terms of the sociocultural and human dynamic of learning in groups, an area that is the focus of my own research and writing around creating culture. It is complex in terms of the neuroscience of learning, a burgeoning field that is both validating and challenging what we know about learning and the best ways to provoke it. Teaching is conceptually complex in terms of the establishment of the curricular goals that form the fundamental content of what gets taught and learned. It is practically complex in terms of establishing the time and structures needed to best carry out those goals. Finally, it is intellectually complex. We see this as schools transition from the transmission model of dispensing facts to be memorized toward a more conceptual model focused on the integration of those facts into rich understandings that help us to comprehend our ever-evolving world.

Concept-Based Curriculum and Instruction for the Thinking Classroom does not skirt any of this complexity; rather it jumps right into the middle of it all with the call to move from a two-dimensional curriculum focused on imparting knowledge and skills to a three-dimensional curriculum aimed at fostering conceptual understanding. Drawing on their three decades of experience in curriculum design, authors Erickson, Lanning, and French serve as our knowledgeable guides in navigating our way through this complex terrain. Using examples from classrooms and integrating research on learning, the authors make a compelling case for the need to organize the curriculum around important, transferable understandings of the discipline in order to add coherence to the collection of skills and knowledge that often make up standards documents and traditional curricular guides. Most important, they champion the necessary role teachers play in this

transformative process.

Based on the individual experience each of us has had in schools and bolstered by the evidence marshaled by Erickson, Lanning, and French, I expect most readers will be convinced of the value of a Concept-Based Curriculum not only for their own students but also for those in schools everywhere. Having made their case, another layer of complexity then emerges for the reader, the practical complexity of actually achieving such a lofty goal.

It is here that the book becomes a valuable resource for readers in embarking on this challenging but important journey of curricular transformation. The ideas and practices that make up this journey are indeed nuanced, connected, and challenging. At times, readers may well wish the authors were beside them as guides offering examples, encouragement, explanation, and probing questions as they work through the process of constructing Concept-Based Curriculum. And indeed, the authors are there through the vignettes they offer from real classrooms, the reflections they share regarding their own learning, the useful templates and practical tips they provide, and the guiding questions they present at the conclusion of each chapter.

As useful as readers will find this resource for all it offers in assistance in planning Concept-Based Curriculum, there is no shying away from the fact that this is a complex journey the authors are encouraging us to take. This book is not about a few superficial ideas and simple suggestions to tack onto your units; it is about full-blown transformation that asks us to shift our understanding of how curriculum is organized and delivered. The authors are also clear that this is not a quick journey; it will take time. In fact, learning to design Concept-Based units is only the first step; a curricular unit only becomes a vehicle for learning when we breathe instructional life into it. And this adds yet another fascinating layer of complexity, the complexity of implantation and instruction, but rest assured your efforts will elevate not only your teaching but also the learning in your classroom and at your school.

—Ron Ritchhart September 6, 2016

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Dedication

Dedicated to teachers the world over who are on the Concept-Based journey to nurture the minds and hearts of young people.

Introduction

Complexity. In a globally interdependent world, the concept of *complexity* frames the sociopolitical and environmental issues of the day. Strongly held beliefs and values lead to increasing polarization and conflict. Competing perspectives and ideologies continuously pull in this and that direction.

Educators are charged with preparing students to live and work in this complex, interactive world. Intuitively, they realize that helping students learn to use their minds well must be a major focus for instruction. But even with this realization, they struggle with the realities of educational challenges:

- How can I develop the critical, creative, and conceptual minds of students and still teach the required content?
- How can I meet the expectations of academic standards given the time constraints and the diversity of student needs?
- How do I design curriculum and instruction to truly develop each student's intellect and problem-solving ability?
- Where can I find the time and professional training to prepare myself for the complexities of twenty-first-century teaching?
- How can I find the time to plan effectively for teaching and learning?

This book provides many answers to the questions—but it requires a mind shift in traditional ways of looking at curriculum design and pedagogy. It requires a shift toward the model we call “Concept-Based Curriculum and Instruction” or CBCI. The insights in this book are grounded in cognitive science, learning theory, and the commonsense reasoning of what works in teaching and learning. Our educational journeys have provided opportunities to teach, evaluate, design, stumble, forge on, and synthesize what works in teaching and learning. For the past three decades, our work has focused intensively on curriculum design and instruction at all levels and in all subjects. This book shares the understandings of curriculum and instruction gleaned from our journeys. These insights are not frivolous fads or fanciful musings. They are deeper insights into the inherent structures of knowledge and process, and into the relationships among curriculum design, instruction, and the development of intelligence.

The following six findings summarize the insights. The first insight is the primary focus of this book:

1. The key to intellectual development is the *synergistic interplay* between the factual and conceptual levels of thinking. Synergy refers to an interaction in which the sum effect is greater than either agent acting alone. Traditional curriculum models generally fail to systematically set up this intellectual syn-

ergy. When curriculum and instruction require students to process factual information and processes through the conceptual level of thinking, the students demonstrate greater retention of their learning, deeper levels of understanding, and increased motivation for their studies.

2. Traditional curriculum design models fail to provide a strong conceptual structure for the information base. The educational result is a *coverage* curriculum model, which encourages cognitively shallow teaching and learning.
3. Every discipline has an inherent conceptual structure. As the information base expands, these conceptual structures become increasingly important for patterning, sorting, and processing incoming information.

The greater the amount of factual information, the greater the need to rise to a higher level of abstraction to organize and process that information.

4. In addition to specific content knowledge and skills, a curriculum needs to articulate clearly the important concepts and conceptual understandings for each grade level and each discipline. These statements are the essential ideas that students must *understand* at a deeper level. The factual knowledge and lower level skills are what students must *know* and be able to *do* in order to describe, discuss, explain, or analyze the deeper concepts. One cannot understand at the conceptual level without knowing the supporting facts and skills. But there must be a synergy between the lower and higher levels of thinking if we are to develop intelligence systematically.
5. The transfer of understanding occurs at the conceptual level. Generalizations and principles supported by facts and skills allow students to see patterns and make connections between new examples of the same concept and conceptual understandings learned previously.
6. Educators often wonder why children enter school as eager, motivated learners but become harder to motivate as they move beyond third grade. Could it be that there is an inverse relationship between an expanding fact base through the grade levels and engagement of the child's personal, conceptual mind? The Concept-Based Curriculum and Instruction model can alleviate this problem by ensuring that the student's conceptual mind remains engaged as the fact base increases through the grade levels.

The early primary grade curriculum is far more conceptual than factual. Teachers invite children to put their minds, hearts, and hands to work in understanding concepts such as colors, weather, family, fairy tales, and numbers. Children bring their personal intellect to the tasks as they collaborate, create, and problem solve. Each child feels the personal exhilaration of using his or her mind well. Joey loves to learn because he is using his own brain. But Joey moves through the grades and something happens. There is a subtle shift from engaging Joey's conceptual mind to covering the growing body of factual content. Almost imperceptibly, at first, Joey begins to lose interest in learning. Teachers think it must be the action video games and sound-bite

television programs that are creating the apathy in classrooms. As the factual and skill loads increase, the conceptual intellectual engagement and consequent motivation for learning decreases in traditional curriculum models. But teachers can fix this design problem with a Concept-Based model for curriculum and instruction.

We live in an age that presents complex problems on both the environmental and sociopolitical fronts. Effective responses to these problems require the abilities to analyze, abstract, conceptualize, predict, collaborate, plan, and act responsibly. Citizens are continually bombarded with persuasive media reports that use words and images to present particular perspectives. Learning how to consider multiple sides of an issue, ask clarifying questions, and challenge biased, incorrect, or inadequately supported arguments are no longer nice skills to have—they are essential skills for twenty-first-century living and problem solving.

What Is New in This Second Edition?

This second edition of *Concept-Based Curriculum and Instruction*, coauthored by Lynn Erickson, Lois Lanning, and Rachel French, is a major revision from the 2007 book. This edition focuses specifically on supporting teachers in grades preschool through high school as they continue their Concept-Based journeys. A significant addition to this book is Dr. Lois Lanning's Structure of Process, which is the complement for the Structure of Knowledge. The Structure of Process speaks for process-driven disciplines: English language arts, the world languages, the visual and performing arts, and music.

The writing of educational leaders and researchers, such as Ron Ritchhart, Linda Elder, Richard Paul, and Carol Tomlinson, support the updated thinking in this book. A detailed set of unit design steps and exemplar units provide teachers with models for moving CBCI into their own classrooms.

A chapter on inquiry learning in lesson planning reinforces the close relationship between instruction that is Concept-Based and learning through inquiry. Inductive teaching engages students in the inquiry process as one of the important strategies to draw out conceptual understandings. Lesson plans are included in the chapter that show how to address the common difficulties teachers encounter when they prepare for inquiry-based teaching and learning.

The final chapter discusses teacher attitudes and beliefs during the change process. Four rubrics developed by Lois Lanning aid teachers in self-assessing their journeys from novice to emerging to master Concept-Based teacher. The rubrics provide clear criteria for assessing "Understanding Concept-Based Curriculum

and Instruction,” “Concept-Based Unit Planning,” “Concept-Based Lesson Planning,” and “Concept-Based Instruction.”

The second edition of this book contains so many new and detailed examples to help teachers read with meaning and put designs into practice. Our own developing understanding helps us explain complex ideas with greater clarity for the reader.

<https://doi.org/10.4135/9781506355382>

Concept-Based Curriculum Glossary of Terms

Concept: A mental construct that frames a set of examples sharing common attributes. Concepts are timeless, universal, abstract (to varying degrees), and have examples that share common attributes. Examples: cycles, diversity, interdependence, inequalities, text evidence, intent, endurance, vanishing point.

Conceptual Lens: A macroconcept (generally) that invites the learner to bring her or his personal intellect to the unit under study. The conceptual lens focuses the direction of the study and creates an intellectual synergy between the lower and higher processing centers in the brain to facilitate deep understanding and transfer.

Deductive Instruction: Instruction that shares a target generalization with students at the beginning of the lesson. Subsequent learning experiences focus on having students identify the facts or skills, often through inquiry, that support or prove the understanding.

Generalization: Two or more concepts stated as a relationship. Also known as *essential understandings*, *enduring understandings*, or *big ideas*. Generalizations are conceptual understandings that transfer through time, across cultures, and across situations. They reflect the deeper, transferrable understandings associated with specific factual content or skills.

Inductive Instruction: Instruction that requires students to examine, categorize, organize, and ask questions about patterns across examples and concepts in order to construct their own generalizations.

Integrated Thinking: The “meta-analysis” of factual information. When thinking is integrated, students see patterns and connections between the facts and generalizations that transfer through time, across cultures, and across situations. A conceptual lens, or conceptual question, facilitates integrated thinking.

Interdisciplinary Curriculum: A curriculum unit in which a variety of disciplines contribute different dimensions of knowledge, skill, and understanding to an issue or problem under study. A conceptual lens ensures the integration of thinking as a result of the interdisciplinary connections.

Intradisciplinary Curriculum: A single discipline curriculum unit with a conceptual lens providing focus for a topic. Thinking becomes integrated at the conceptual level as learners reach conceptual understanding supported by facts or skills.

Macroconcepts: The broadest, most abstract concepts that transfer across subject areas or organize more specific microconcepts within disciplines. Macroconcepts are often used as conceptual lenses for units of study to develop *breadth* of knowledge in inter- and intradisciplinary contexts. Examples: interdependence, change, systems, power, intent, movement.

Microconcepts or Subconcepts: The more specific concepts in a discipline. Students internalize these concepts, as well as their expanding relationships, through the grade levels to develop disciplinary *depth*. Examples: organism, proportion, migration, habitat, scarcity, mythology, voice.

Principles: A generalization that is considered a foundational truth for a discipline. A principle does not use qualifiers, e.g., scientific laws (Newton's First Law of Motion)

Structure of Knowledge: A visual representation between, on the one hand, topics and facts and, on the other, the concepts, generalizations or principles, and theories that are drawn from and supported by the factual foundation. (See Chapter 2.)

Structure of Process: A visual representation of the relationship of processes (and the components of a process) to concepts, generalizations or principles, and theories. (See Chapter 2.) Concepts are drawn from and supported by the essential processes, strategies, and skills found within the unit. Lanning (2013) further defines the following components of the Structure of Process:

- Processes are actions that produce results. A process is continuous and moves through stages during which inputs (materials, information, people's advice, time, etc.) may transform or change the way a process flows. A process defines what is to be done—for example, the writing process, the comprehension process, the research process, the respiratory process, and so on.
- Strategies may be thought of as the systematic plans learners consciously adapt and monitor to improve their learning performance. Strategies are complex because many skills are situated within a strategy.
- Skills are the smaller operations or actions that are embedded in strategies and, when appropriately applied, “allow” the strategies to work.

Synergistic Thinking: A cognitive interplay between the factual and conceptual levels of thinking leading to deeper, transferable understanding and increased motivation for learning.

Theory: A supposition or set of conceptual ideas used to explain a phenomenon or practice.

Concept-Based and IB Terminology Correlations

Concept-Based	IB
1. Macroconcepts (broad, transfer across subjects)	1. Key Concepts
2. Microconcepts (generally discipline specific)	2. Related Concepts (macroconcepts can sometimes be used as related concepts if applied to a specific discipline)
3. Conceptual Lens (one or two macroconcepts used to focus a unit of study, emphasize conceptual transfer, and engage synergistic thinking)	3. Key Concepts (used to focus a unit of study and emphasize conceptual transfer across situations)
4. Generalizations or Principles (conceptual understandings that show important conceptual relationships; deeper understandings that transfer through time, across cultures, and across situations)	4. Central Ideas; Statements of Inquiry
5. Guiding Questions (factual, conceptual, and debatable)	6. Teacher Questions (The MYP uses the factual, conceptual, and debatable questions)
7. Synergistic Thinking (the intellectual interplay between the factual or skill and conceptual levels of thinking, which engages the personal intellect and is the key to intellectual development)	IB does not have this term, but the IB design effects synergistic thinking.
8. Two-Dimensional Versus Three-Dimensional Curriculum and Instruction (2D = the coverage model of facts and skills; 3D teaches facts and skills but requires students to understand them in relation to important concepts and conceptual understandings)	IB does not use this term, but IB, by design, is a three-dimensional curriculum and instruction model.

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Sample Verbs for Level 2 and 3 Generalizations

Accelerate

Accomplish

Achieve

Acquire

Activate

Adapt

Address

Adjust

Administer

Advance

Allocate

Analyze

Anticipate

Approve

Arrange

Ascertain

Assemble

Assess

Assign

Assimilate

Assist

Assure

Attain

Attend

Balance

Bring

Bring about

Build

Calculate

Challenge

Chart

Check

Clarify

Classify

Collect

Command

Communicate

Compile

Compose

Compute

Conceive

Condense

Conduct

Conserve

Consolidate

Construct

Contribute

Convert

Convey

Cooperate

Coordinate

Correlate

Correspond

Create

Cultivate

Deal

Decide

Define

Delegate

Deliver

Demonstrate

Design

Detect

Determine

Develop

Devise

Direct

Discover

Display

Distribute

Document

Draft

Dramatize

Draw

Draw up

Drive

Earn

Edit

Elaborate

Eliminate

Empathize

Employ

Enact

Encourage

Enforce

Engineer

Enhance

Enlist

Ensure

Equip

Establish

Estimate

Evaluate

Examine

Execute

Expand

Expedite

Experiment

Explain

Express

Facilitate

Fashion

Finance

Fix

Follow

Forecast

Forge

Form

Formulate

Function as

Gain

Gather

Generate

Give

Govern

Guide

Handle

Help

Hypothesize

Identify

Illustrate

Imagine

Implement

Improve on

Improvise

Increase

Inform

Initiate

Innovate

Inspect

Inspire

Install

Instill

Institute

Instruct

Insure (ensure)

Integrate

Interpret

Introduce

Invent

Investigate

Judge

Justify

Keep

Kindle

Launch

Lead

Learn

Lift

Listen to

Locate

Maintain

Make

Manage

Manipulate

Market

Master

Mediate

Meet

Memorize

Mentor

Minimize

Model

Monitor

Motivate

Move

Navigate

Negotiate

Nominate

Observe

Obtain

Offer

Optimize

Orchestrate

Order

Organize

Originate

Overcome

Oversee

Paint

Participate

Perceive

Perfect

Perform

Persuade

Photograph

Pilot

Pioneer

Place

Plan

Play

Predict

Prepare

Prescribe

Present

Prevent

Problem Solve

Process

Procure

Produce

Project

Promote

Propose

Protect

Prove

Provide

Publicize

Purchase

Question

Raise

Read

Realize

Reason

Receive

Recognize

Recommend

Reconcile

Record

Recruit

Rectify

Redesign

Reduce

Reevaluate

Refer

Refine

Regulate

Relate

Remember

Render

Reorganize

Repair

Report

Represent

Research

Resolve

Respond

Restore

Retrieve

Revamp

Review

Revise

Revitalize

Revive

Rout

Save

Schedule

Screen

Secure

Select

Sell

Sense

Separate

Serve

Service

Set up

Shape

Share

Shift

Show

Simplify

Sketch

Solidify

Solve

Sort

Spark

Speak

Spearhead

Staff

Stimulate

Streamline

Strengthen

Stress

Stretch

Structure

Study

Substitute

Succeed

Summarize

Supersede

Supervise

Supply

Survey

Symbolize

Synergize

Talk

Teach

Tell

Tend

Test

Trace

Track

Train

Transcribe

Transfer

Transform

Translate

Travel

Treat

Trim

Uncover

Undertake

Unify

Unite

Update

Upgrade

Use

Utilize

Validate

Verify

Widen

Win

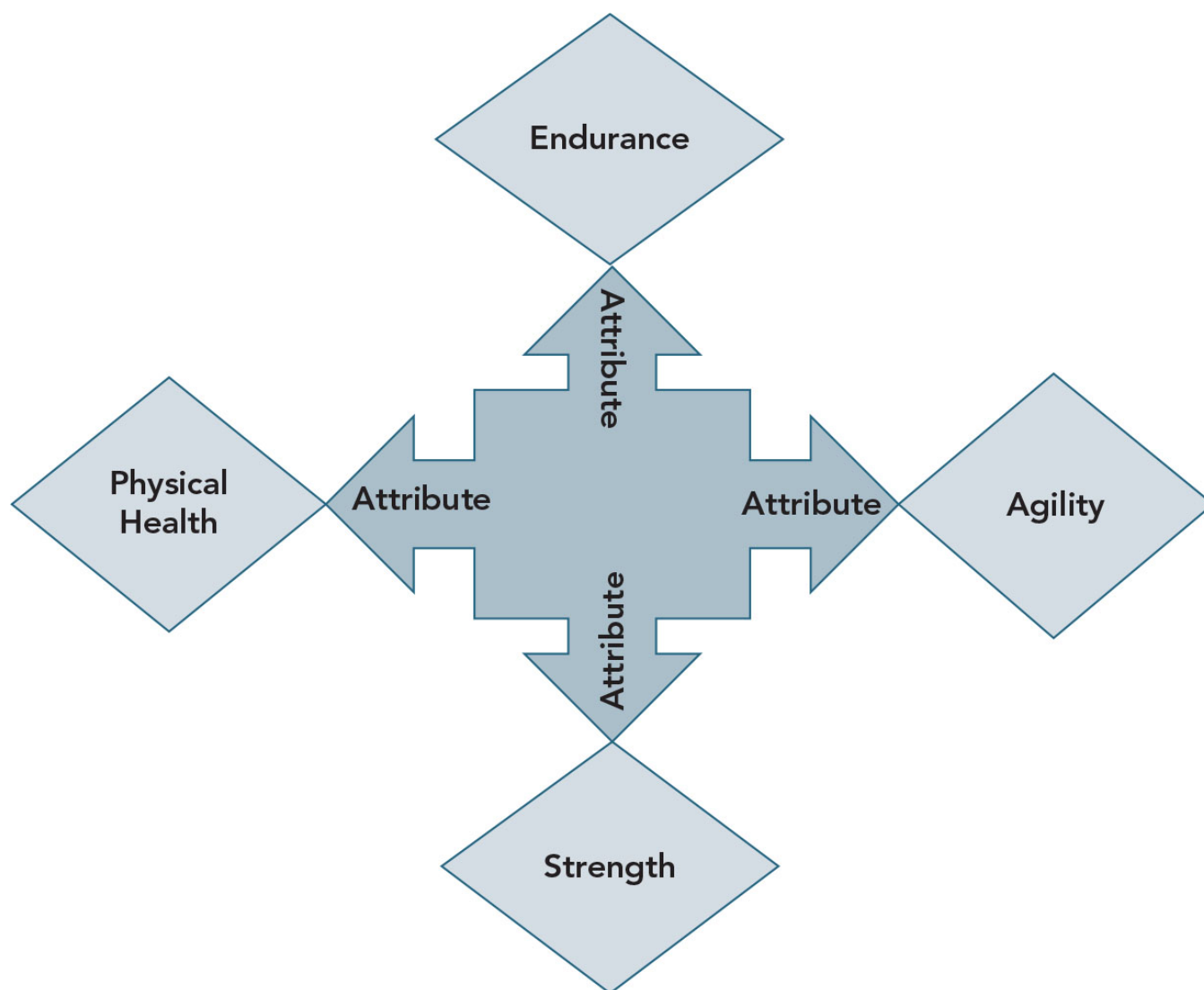
Withdraw

Work

Write

Concept-Based Graphic Organizers

Figure C1: Concept graphic organizer



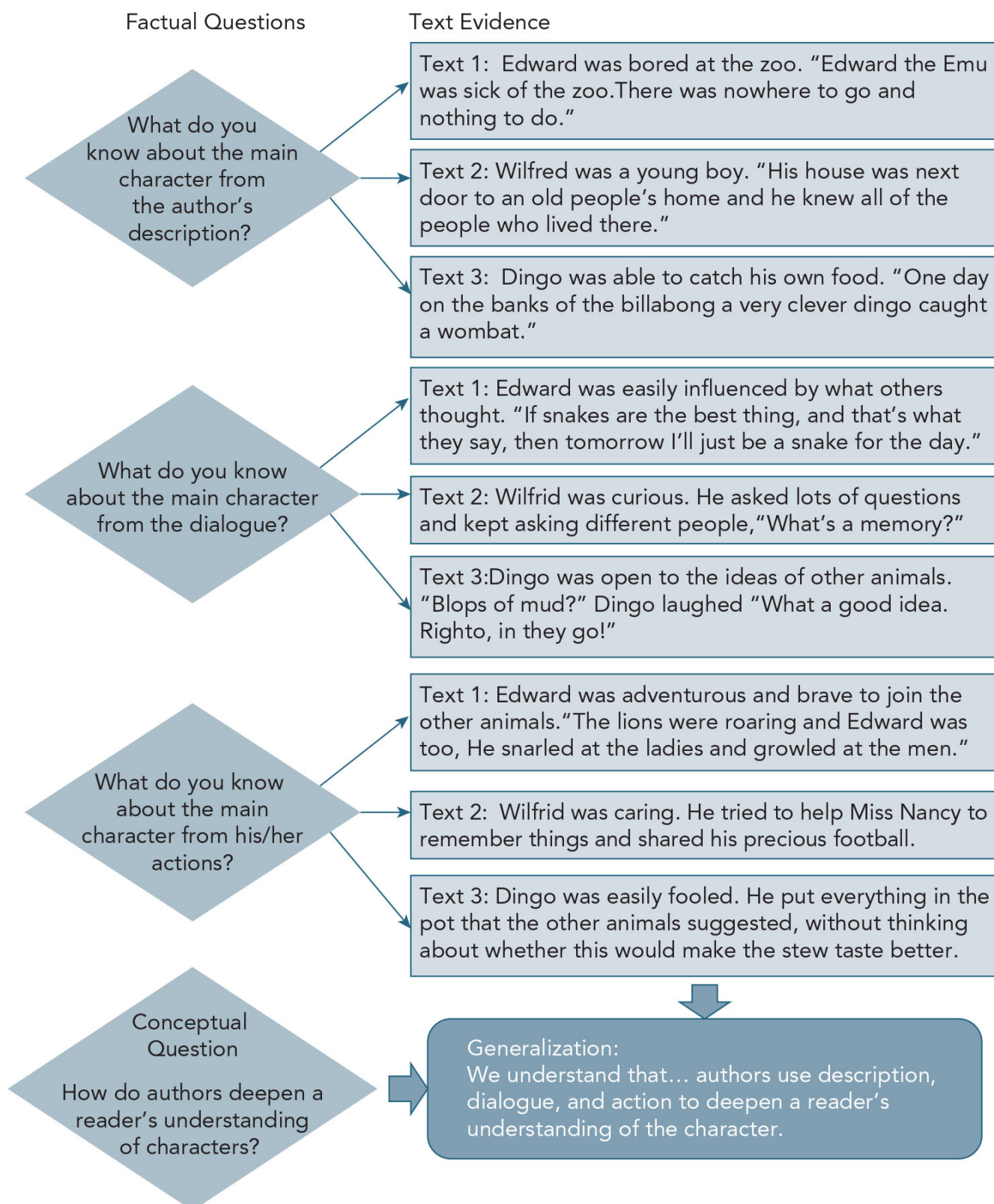
Consider these defining attributes. What concept could they collectively represent?

Figure C2: Painting Graphic Organizer for Visual Art

Drawing	Perspective What is the viewer's perspective in each of these drawings?	Line How do the artists of these drawings use line to create perspective?	Shape How do the artists of these drawings use shape to create perspective?	Generalizations Infer an understanding about the use of lines and shapes to convey perspective in landscape drawings based on your observations. We understand that...
Landscape Drawing #1 (a distance view)	The viewer is looking into the distance.	The artist used converging lines to create a vanishing point, suggesting a distance perspective.	The foreground has larger shapes. Shapes in the background are smaller to create a distance perspective.	Artists use converging lines and larger and smaller foreground and background shapes to create a distance perspective.
Landscape Drawing #2 (a close view)				
Landscape Drawing #2 (a view from above)				

Note: This example uses three landscape drawings and asks students to consider the viewer's perspective. Students then observe and record how the artists used the elements of line and shape to create perspective. The observations culminate in the students creating generalizations that relate line and shape to perspective.

Figure C3:



Book Talk

When using this graphic organizer, the concepts can be drawn from the content or from the writer's or reader's craft. Generalizations may come from the *Structure of Knowledge* or the *Structure of Process* but should not try to blend both together.

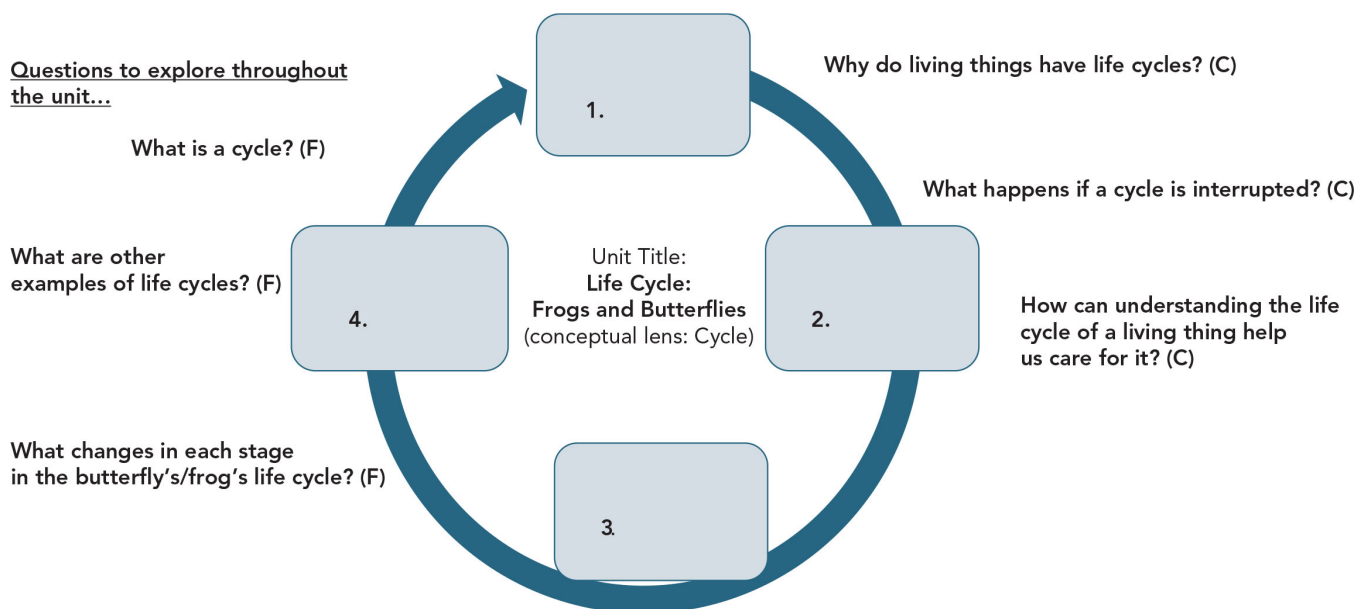
Conceptual Focus: Character Development

Text 1: *Edward The Emu* by Sheena Knowles (1988) Harper Collins

Text 2: *Wilfrid Gordon McDonald Partridge* by Mem Fox (1984) Puffin Books

Text 3: *Wombat Stew* by Pamela Lofts (1984) Ashton Scholastic

Figure C4: Class Organizer for Unit



Complete this “Big Idea” about life cycles: As a result of our learning about the life cycles of frogs and butterflies, we understand that....**life cycles**

Concept-Based Curriculum: Unit Design Steps

Lynn Erickson and Lois Lanning

Step 1: Create the unit title.

(The unit title can be engaging for students but needs to clearly indicate the content focus.)

Step 2: Identify the conceptual lens.

The conceptual lens is a concept that provides focus and depth to the study and ensures synergistic thinking (connecting both the factual and conceptual).

Step 3: Identify the unit strands.

Strands will be subject areas for interdisciplinary units. For intradisciplinary units, the strands will be major headings, which break the unit title into manageable, instructional parts. In an intradisciplinary process subject area unit, the strands are predefined: *understanding, responding, critiquing, and producing*. In a process discipline, the strands are defined: understanding, responding, critiquing, and producing.

Strands are placed in a web around the unit title.

Step 4: Web out the unit's topics and concepts under the strands.

After brainstorming, underline the concepts under each strand, so they can be easily accessed in the next step.

Step 5: Write the generalizations that you expect students to derive from the unit study.

You will craft one or two generalizations using the conceptual lens—and one or two generalizations for *each* of the strands. Sometimes, a generalization will address one or more strands (especially in a process discipline). A unit of study may have five to nine generalizations depending on the grade level and length of time to be spent on the unit.

Step 6: Brainstorm the guiding questions.

Guiding questions facilitate student thinking toward the generalizations.

Guiding questions should be coded as to type (factual, conceptual, or debatable). Each generalization needs a mixed set of three to five factual and conceptual questions, developed during the planning process, and two

to three debatable questions for the unit as a whole.

Step 7: Identify the critical content.

The critical content is the factual knowledge required for grounding the generalizations, deepening knowledge of the unit topic, and defining what students may need to know about processes or skills.

Step 8: Identify the key skills.

The key skills are drawn verbatim from academic standards or national curricula.

Key skills transfer across applications and are not tied to specific topics until they appear in the learning experiences.

Step 9: Write the common, culminating assessment and scoring guide or rubric.

The culminating assessment reveals student understanding of one or two important generalizations, their knowledge of critical content, and their ability to apply key skills. Develop a scoring guide, or rubric, with specific criteria for evaluating student work on the task.

Step 10: Design suggested learning experiences.

Learning experiences ensure students are prepared for the expectations of the culminating assessment and reflect what students should understand, know, and be able to do by the end of the unit. Learning experiences are meaningful and authentic. Included in this section are suggestions for pacing, other assessments, differentiation strategies, and unit resources.

Step 11: Write the unit overview.

The unit overview is written for the teacher to read to the students as a hook to grab their interest and attention and to introduce them to the study.

Concept-Based Curriculum: Unit Templates

Grade Level: _____

Unit Number: _____

Title and Conceptual Lens: _____

Unit Length _____

Author(s): _____

Unit Overview:

Technology Integration:

Standards or national curricula addressed in this unit:

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Unit Title _____

Conceptual Lens _____ Grade Level _____

Strand 1

Strand 2

Unit Title

Strand 3

Strand 4

*Add strands as needed

Grade Level _____ Unit Title _____

Conceptual Lens _____

Generalizations

Guiding Questions: F = Factual; C = Conceptual; D = Debatable

*Add cells as needed

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Critical Content and Key Skills

**Critical Content
by Strands****Key Skills**

Students will know . . .

Students will be able to . . .

Strand 1

Strand 2

Strand 3

Strand 4

* Some key skills may be correlated 1-1 to the “knows” in process-driven subjects; but they are less likely to

be correlated 1-1 in content subjects such as history and science.

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Unit Title _____

Grade Level _____

Suggested Time Line	Suggested Learning Experiences	Assessments	Differentiation	Resources

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Culminating Unit Assessment

Unit Title _____

Grade Level _____

What (the unit focus):

Why (the generalization or generalizations):

How (the engaging scenario for students):

Scoring Guide

For Letter Grades

	Task:	Possible Pts. or %	Self-Assess.	Teacher-Assess.
STANDARD	Scoring Criteria			
	<i>Understanding:</i>			
	<i>Content:</i>			
	<i>Process:</i>			
		100	_____	_____
Scoring Key				
A =				
B =				
C =				
I =				

Scoring Guide

Primary Grades

4 Meets all standard criteria plus....

Examples:

S T A N D A R D	Understanding	✓ If Met
		☐
	Content	☐
		☐
	Process	☐
		☐

2 Meets 3/5 of the criteria in the standard

1 Not able to score; does not yet approach standard

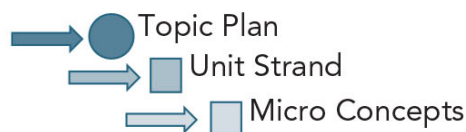
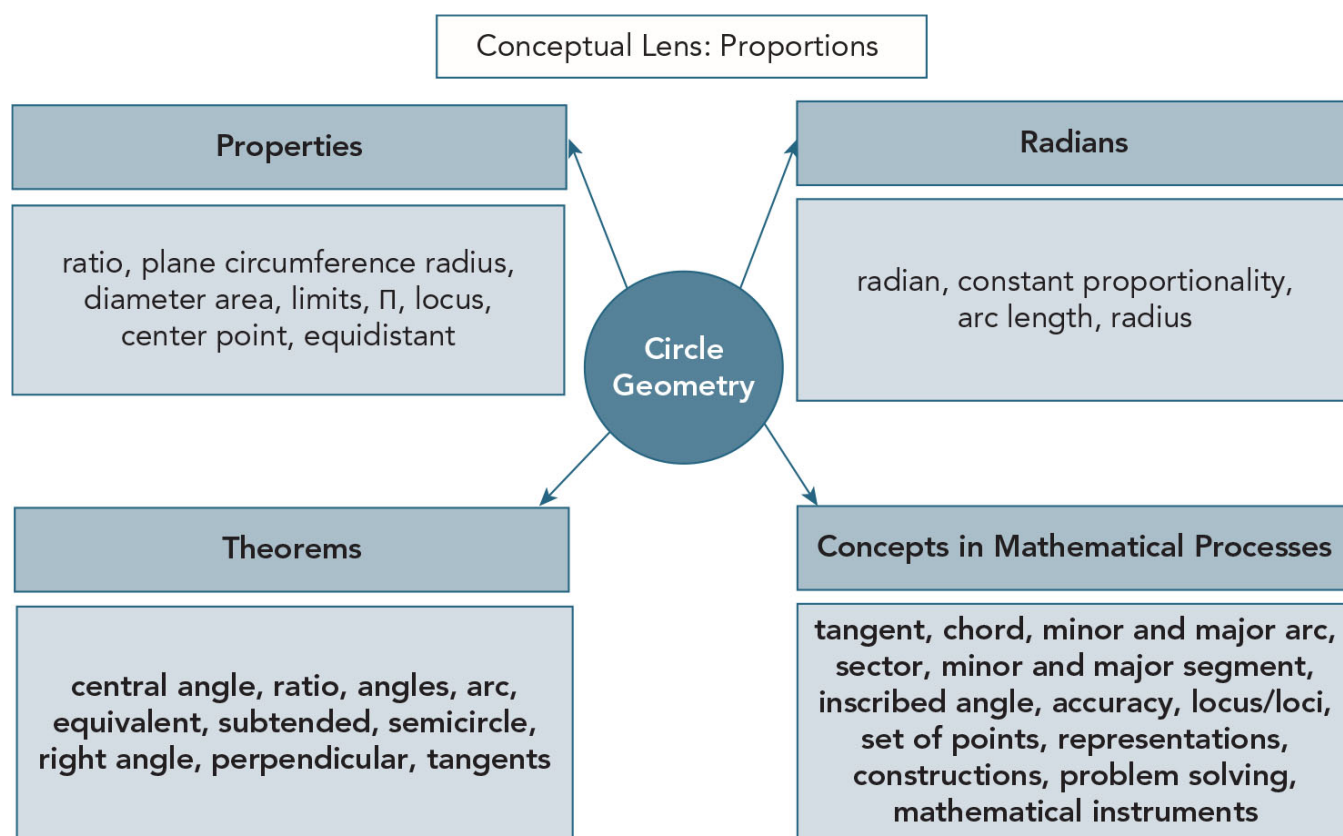
Note: Adjust standard 2 to correspond to the number of scoring boxes.

Concept-Based Curriculum: Unit Examples and Culminating Unit Assessments

- *Resource D-3.1 Circle Geometry*
- *Resource D-3.2 Are We Really Surrounded by Waves?*
- *Resource D-3.3 Building Language Through Play: Expression (Early Years)*
- *Resource D-3.4 English Language Arts*

Unit Example: Circle Geometry

Circle Geometry Unit Web



Source: Jennifer Chang Wathall © 2015 *Concept-Based Curriculum and Instruction for Secondary Mathematics*, Thousand Oaks, CA: Corwin.

Unit Title: Circle Geometry	Conceptual Lens: Proportions	Time Allocation: 4–6 weeks	Grade/Year: Grade 8/Year 9
Unit overview: Circles fascinated people for many years. How often do you see a circle in real life? Circles are often used to represent unity and harmony. How would you construct some of the circle designs in ancient artifacts? http://nrich.maths.org/2561	Concepts in the unit: ratio, plane, circumference, radius, diameter, area, limits, π , locus, center point, equidistant, central angle, ratio, angles, arc, equivalence, angle subtended, semicircle, right angle, perpendicular, radian, constant proportionality, arc length, chord, minor and major arc, sector, minor and major segment, inscribed angle, accuracy, locus/loci, set of points Process concepts: representations, constructions, problem solving, mathematical instruments		
What we would like students to know	What we would like students to understand	What we would like students to be able to do	
1. Know the formula for the area of circumference and area of a circle. 2. Define the radian measure of the angle as the constant of proportionality. 3. Know circle vocabulary: circumference, radius, diameter, chord, minor and major segment, tangent, minor and major arc, locus, right angle, arc length.	1. The ratio of circumference to diameter in all circles represents a fixed constant, π . 2. The locus of points equidistant from a center point on a plane corresponds to circumference. 3. Summing the limit of sectors forms a rectangle, which formulates areas for circles. 4. Angles inscribed in semicircles correspond to right angles. 5. Radii and tangents form right angles (perpendicular angles). 6. Central angles represent double the inscribed angle. 7. Angles subtended from the same arc convey equivalence. 8. The radian measure of angles represents constant proportionality between the radius and arc length. 9. Radian measure simplifies formulae for arc lengths and areas of sectors. 10. Knowing how to draw circle parts accurately (e.g., tangents, chords, segments, sectors, arcs, and angles) helps in the solution of geometry problems. 11. The correct use of mathematical instruments to construct different loci demonstrates the shared properties that govern a set of points.	<i>Process: Making connections</i> 1. Convert radians to degrees and vice versa. <i>Process: Reasoning and proof</i> 2. Prove all circles are similar. 3. Derive, using similarity, the fact that the length of the arc intercepted by an angle is proportional to the radius. 4. Derive the formula for the area of a sector. <i>Process: Creating representations and reasoning and proof</i> 5. Construct the inscribed and circumscribed circles of a triangle, and prove properties of angles for a quadrilateral inscribed in a circle. 6. Construct a tangent line from a point outside a given circle to the circle.	
Generalizations (Students understand that . . .)		Guiding Questions	
1. The ratio of circumference to diameter in all circles represents a fixed constant (π).		Factual Questions: What is the formula to find the circumference and area of a circle? What do the following words mean? circumference, radius, diameter, chord, minor and major segment, tangent, minor and major arc, locus, right angle, arc length Conceptual Questions: How are all circles similar? Why does the ratio of circumference to diameter in all circles represent a fixed constant? How does the ratio of circumference relate to the diameter in all circles? Factual Questions: What is the definition of circumference? What does <i>locus of points</i> mean? Conceptual Questions: How does the locus of a set of points explain circumference?	
2. The locus of points equidistant from a center point on a plane corresponds to circumference.		Factual Questions: What is the formula for the area of a circle? Conceptual Questions: How does one find the area of a circle by using the limiting sum? How is a circle similar to a rectangle?	
3. Summing the limit of sectors forms a rectangle, which formulates areas for circles.		Factual Questions: What are the circle theorems? Conceptual Questions: How do you identify and describe relationships among inscribed angles, radii, and chords?	
4. Angles inscribed in semicircles correspond to right angles. 5. Radii and tangents form right angles (perpendicular angles). 6. Central angles represent double the inscribed angle. 7. Angles subtended from the same arc convey equivalence.		Factual Questions: What is the definition of a radian? Conceptual Questions: Why are radians dimensionless? How is radian measure represented? How does the concept of “constant proportionality” relate to the radian measure of angles?	
8. The radian measure of angles represents constant proportionality between the radius and arc length. 9. Radian measure simplifies formulae for arc lengths and areas of sectors. Arc lengths and areas of sectors intercepted by the angle relate to the radius.		Factual Questions: What is a scale diagram? Conceptual Questions: How do you use constructions to solve problems? Why are constructions useful in the problem-solving process? When is the accurate drawing of circle parts critical for problem solving?	
10. Accurate geometry solutions depend on exact drawings of circle parts, such as tangents, chords, segments, sectors, arcs, and angles, when using drawing as problem solving.		Factual Questions: What are mathematical instruments and applets? Conceptual Questions: How do you use construction and applets to demonstrate shared properties that govern a set of points? Why are applets useful in circle geometry?	
11. The correct use of mathematical instruments to construct different loci demonstrates the shared properties that govern a set of points.			
Debatable Unit Questions			
Which is better for measuring angles: radians or degrees?			

Culminating Unit Assessment: Circle Geometry

Unit Title: Circle Geometry

Grade Level: 8

What (the unit focus): Investigate circle theorems ...

Why in order to understand that ...

- Angles inscribed in semicircles correspond to right angles.
- Radii and tangents form right angles (perpendicular angles).
- Central angles represent double the inscribed angle.
- Angles subtended from the same arc convey equivalence. *

How (the engaging scenario for students):

In groups of four and as documentary film makers, create a documentary on the four circle theorems, which include angles in semicircles, the relationship between tangents and radii, central angles and angles subtended at the circumference, and angles subtended from the same arc. Build your own GeoGebra applets for each circle theorem, and use three numerical values to form your generalization for each circle theorem. Illustrate each theorem with diagrams, explanations, examples, and how to apply each theorem.

* Note: Typically a culminating assessment task has no more than two generalizations. This task is exempt from this rule because the generalizations are highly interdependent.

Scoring Guide

For Letter Grades

Task: Documentary Film

Scoring Criteria

Possible Pts. or %

Self-Assess.

Teacher-Assess.

S
T
A
N
D
A
R
D

Understanding:

Your generalizations for each theorem were accurate, demonstrating understanding.

20%

The examples of how to apply the theorems showed the ability to transfer understanding.

20%

Content:

Your video provided clear explanations of each circle theorem in terms of the factual content.

20%

You provided three numerical examples.

Process:

You collaborated in a harmonious fashion with your group.

10%

You built the GeoGebra applets successfully for each theorem.

10%

Your video presentation was well crafted:

- diagrams, explanations, and examples were clear, logically presented, and held viewer interest.

10%

10%

100%

Scoring Key

A =

B =

C =

I =

Unit Example: Are We Really Surrounded by Waves?

Grade Level: 1st

Title: Are We Really Surrounded by Waves? Understanding Sound and Light

Conceptual Lens: Cause and Effect

Unit Length 6-8 Weeks

Author(s): Erica Tolf, Kaylee Panek, the Burlington-Edison School District First Grade Team and Amy Reisner, Vice Principal, Bay View Elementary, Burlington, Washington

Unit Overview:

Do you know that we are surrounded by waves? Not ocean waves; not goodbye waves ... we are surrounded by invisible waves! We know they are there because they create light and produce sound.

In this unit, we will learn about these amazing, invisible waves and how we can use them to communicate with each other.

At the end of this unit, you will be able to design an ambulance that can communicate emergency messages to people who may need help.

Technology Integration:

iPad apps: Mooncast, sunrise and fall, pic collage

Standards or national curricula addressed in this unit:

Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate (1-PS4-1).

Make observations to construct an evidence-based account that objects can be seen only when illuminated (1-PS4-2).

Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light (1-PS4-3).

Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance (1-PS4-4).

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Unit Title: Are We Really Surrounded by Waves?

Conceptual Lens: Cause/Effect Grade Level 1

Strand 1

Properties of Waves

- Sound
- Vibration
- Property
- Length
- Duration

Strand 2

Waves and Light

- Light
- Illumination
- Objects
- Distance

Unit Title

Are We Really
Surrounded by Waves?
Understanding Sound
and Light

Strand 3

Waves and Matter

- Translucent opaque
- Transparent reflect
- Energy
- Property
- Objects
- Matter
- Light

Strand 4

Waves and Communication

- Distance
- Communication
- Sound
- Light
- Receptor
- Generator
- Material

*Add strands as needed

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Grade Level 1

Unit Title: Are We Really Surrounded by Waves? Understanding Sound and Light

Conceptual Lens: Cause/Effect

Guiding Questions:

Generalizations

F = Factual; C = Conceptual;

D = Debatable

Sound waves create vibrations.

What are different sources of sounds? (F) What sounds do different things make? (F) How are sounds made? (C)

How do vibrations of different materials cause sounds to change? (C)

How do sounds cause things to vibrate (move)? (C)

Would you rather live in a world without sound or loud sounds all the time? Explain. (D)

Light comes from different sources.

What are different light sources? (F; natural light vs. human-made light)

Why are light sources important or useful? (C) What is illumination? (F)

What would happen if a light source is not available? (C)

What are some of the benefits of light from different sources? (C)

What if we couldn't ever turn off the lights? (D)

How can we conserve our source of light?

Different materials direct how light waves travel.

How does light travel? (C)

What makes a shadow? (F)

Why do some objects cast a shadow while others do not? (C)

How do properties of different matter (transparent, translucent, opaque, reflective) affect light travel?

	(C; extension: sonar, satellites, echo location, microwave, radio waves)
	How can light be redirected? (F)
	What are different ways humans communicate? (F)
Sound and light waves assist communications across distances.	What are different ways animals communicate? (F)
	How can sound and light be used to communicate over distances? (C)
	Do you think everyone should be able to use the same lights and sounds to communicate? (D)

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Critical Content and Key Skills

Critical Content by Strands	Key Skills
Students will know ...	Students will be able to ...
Strand 1	1-PS4-1: Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.
How sound is made	1-PS4-2: Make observations to construct an evidence-based account that objects can be seen only when illuminated.
How waves cause vibrations	1-PS4-3: Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light.
Sound sources (instruments, animals, door, car, wind, mouth ...)	1-PS4-4: Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance.
Sound differentiation by pitch and volume	Construct meaning from oral presentations and literary and informational text through grade-appropriate listening, reading, and viewing.
Vocabulary: vibration, wave	Participate in grade-appropriate oral and written exchanges of information, ideas, and analyses, responding to peer, audience, or reader comments and questions.
Strand 2	Speak and write about grade-appropriate complex literary and informational texts and topics.
How illumination works	Construct grade-appropriate oral and written claims and support them with reasoning and
Human-made light vs. natural light	
Vocabulary: illuminate, human-made light sources, natural light source	

Strand 3

Properties of light beams

Properties of light travel

Mirroring

Vocabulary: transparent, translucent,
opaque, reflective

evidence.

Strand 4

Communication using sound and light

Technology using light

Engineering using light

Emergency sounds and communication

Vocabulary: communicate, engineer,
device

Conduct research and evaluate and communicate findings to answer questions or solve problems.

Determine the meaning of words and phrases in oral presentations and literary and informational text.

Unit Title WavesGrade Level 1

Suggested Time Line	Suggested Learning Experiences	Assessments	Differentiation	Resources
	1. Sound sources and vibrations	- Inquiry Chart: What do you know? What did you hear?	- Partners, with adult support.	Listening Walk by Paul Showers
	- Listening walks	- Show a picture file card.	- Use sound journals to record different sounds using pictures or words	What is Sound? By Charlotte Guillain
	- Observation charts with sound pictures	Student writes the sound it makes and points to where it vibrates.		Oscar and the Bat: A Book about Sound by Geoff Waring
	- Sort and compare picture file cards of different instruments	- Give students picture file cards to sort and label as human-made or natural light	Students can use pictures and/or words to show their	Sound and light picture file cards (laminated)
				- Ask parents for donations for creating sound devices: tubes, boxes,

- Create instruments

- Sound vibration stations

2. Exploring light sources and illumination

- Create pic collages of light sources using iPads

- Sort and compare picture file cards of light sources

- Make posters in small groups of light sources and why they are important

- Investigate pinhole boxes and illumination

3. Exploring light and different matter

- Explore with flashlights and different materials

- Sort different materials according to attributes

- Explore and compare shadows

4. Communicating with light or sound

- Cup/string telephones

- Emergency vehicles

source

- Use picture file cards to match with correct vocabulary

- Design and create a device that can be used to communicate over a distance

understanding

- Use pictures to post next to vocabulary words on class chart

rubber bands, shakers, craft sticks, cans or jars

- Make pinhole boxes

- Use flashlights and variety of materials to explore around the room

String and cups

Sound All Around by Wendy Pfeffer

Sound: Loud, Soft, High and Low by Natalie Rosinsky

Culminating Unit Assessment

Unit Title Are We Really Surrounded by Waves? Understanding Sound and Light

Grade Level 1

What (the unit focus):

Students will create an ambulance that can use light and sound to communicate...

Why (the generalizations):

...in order to understand that sound and light waves assist communications across distances.

How (the engaging scenario for students):

You are an ambulance driver.

You will be designing an ambulance that can use light and sound to communicate to drivers.

You need to design lights and sounds that will communicate what you want to drivers.

You need to use different volumes, pitches, and lights.

Please draw this ambulance on a sheet of white paper, using color and detail to explain your ideas.

You will need to be prepared to explain the parts of your ambulance, what you want to communicate to drivers, and how your sounds and lights communicated your messages using waves.

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Scoring Guide

Primary Grades

Meets and exceeds standard criteria:

Meets all standard criteria plus...

4 Examples: (Student's design is creative and innovative increasing clarity of communication; student combines light and sound with communication.)

		✓ If Met
3	S T A N D A R D	Understanding
		Design shows communication across distance.
	Students can explain how light and sound waves communicate and how they work.	
	Content	Design has communication using light.
		Design has communication using sound.
		Pitch is referenced.
		Volume is referenced.
	Process	Reasoning about design is present.
		Communication about design is clear.
		Ambulance drawing shows craftsmanship and effort.

2 Meets 2/3 of the criteria in the standard

1 Not able to score; does not yet approach standard

Unit Example: Building Language Through Play

Expression (Early Years)

Grade Level:

Early Years

Unit Number:

Title and Conceptual Lens:

Building language through play: Expression

Unit Length:

Author(s):

Kacey Mayer from Cham, Switzerland, and Pamela Ostler, from Zürich, Switzerland

Unit Overview:

The child is made of one hundred.

The child has a hundred languages

a hundred hands

a hundred thoughts

a hundred ways

of thinking, of playing, of speaking ...

—Loris Malaguzzi

We all learn through play. In this unit, we will explore how we can express ourselves and how we understand and respond to those around us. Can you look at someone and know what his or her feelings are? Do you know how to express your feelings? Can you resolve conflicts and cooperate with others? Soon we will have answers for all these questions!

Standards or national curricula addressed in this unit:

International Baccalaureate Primary Years Programme Scope and Sequences

PSPE - Interactions - Phase 1

- Identify when their action have impacted others
- Reach out for help when it is needed for themselves or others
- Share their own relevant ideas and feelings in an appropriate manner
- Enjoy interacting, playing, and engaging with others

PSPE Interactions - Phase 2

- Understand the impact of their actions on each other and the environment
- Seek adult support in situations of conflict
- Assume responsibility for a role in a group
- Ask questions and express wonderings
- Discuss and set goals for group interactions

Visual Language - Viewing and Presenting - Phase 1

- Use body language to communicate and to convey understanding, for example, pointing, gesturing, facial expressions
- Interact effectively with peers and adults in familiar social settings
- Understand simple questions and respond with actions or words
- Realize that word order can change from one language to another
- Use own grammar style as part of the process of developing grammatical awareness

Visual Language - Viewing and Presenting - Phase 2

- Use body language in mime and role play to communicate ideas and feeling visually
- Use language to address their needs and express feelings and opinions
- Ask questions to gain information and respond to inquiries directed to themselves or the class
- Use oral language to communicate during classroom activities, conversations, and imaginative play

**Understanding
Language**

Vocabulary/new words
Active listening
Visual cues
Turn taking
Roles within a group
Conflict resolution
techniques
Question and answer
techniques

**Responding to
Language**

Active listening
Body language
Gestures
Eye contact
Cooperation
Verbal and nonverbal
Communication
Turn taking

Unit Title**Building Language
Through Play****Producing Language**

Intonation
Clarity
Vocabulary/new words
Sentence structure
Conversation strategies—
interruptions

Critiquing Language

Voice—mood and tone
Intonation
Body language
Visual cues

Unit Title: Building Language Through Play

Generalizations	Guiding Questions: F = Factual C = Conceptual D = Debatable
1 - Our actions, tone of voice, and words express our feelings and ideas.	1 a. What are feelings? (F) 1 b. Can you tell me about a time you were very happy or very sad? (F) 1 c. What are examples of different tones of voice? (F) 1 d. How do you think XXX was feeling from his tone of voice? From his actions? (F) 1 e. How can you express your feelings and ideas? (C) 1 f. How are our feelings connected to our words and actions? (C) 1 g. Why should we be aware of our tone of voice? (C) 1 h. Can the tone of my voice show more than one emotion? (D)
2 -Cooperation helps us work together and solve conflicts (UL, RL, PL).	2 a. How do we cooperate? (F) 2 b. What is a group experience? (F) 2 c. How can you ask for help? (F) 2 d. Who can you go to if you need help? (F) 2 e. How does cooperation make our experiences better? (C) 2 f. How does turn taking help us cooperate? (C) 2 g. What are some ways that you can solve conflicts by yourself? (C)
3 - Questions and answers help us to gain new information and understandings (UL, RL).	3 a. What are questions? (F) 3 b. What are answers? (F) 3 c. Are all questions the same? Can you give me some examples of questions you have asked today? (F) 3 d. How can questions help us find out new information? (C) 3 e. Why do we need to listen to the answers given to our questions? (C)

	3 f. How can the answers we are given help us to learn new things? (C)
	4 a. How do we listen with our body? What is active listening? (F)
	4 b. What is body language? (F)
	4 c. How do moods affect body language? (C)
4 - Our body language expresses our feelings (PL, RL).	4 d. How does active listening help us understand each other? Can we listen to each other when more than one person is speaking? (C)
	4 e. Can you express how you are feeling by only using body language? (C)
	4 f. What is the difference between open and closed body language? (F)
	5 a. What is a sentence? (F)
	5 b. What is a message? (F)
5 - The order of our words shapes our message (PL).	5 c. What is a question? (F)
	5 d. What is a statement? (F)
	5 e. Does changing the order of your words change the meaning of your message? (C) Why or why not? Can you give me an example? (F)

Critical Content and Key Skills

Critical Content by Strands	Key Skills
Students will know...	Students will be able to...
Understanding Language	Understanding Language
• Roles within a group • Conflict resolution techniques • Question and answer techniques	• Ask questions to learn from others • Seek adult support in situations of conflict • Assume responsibility for a role in a group

Responding to Language		Responding to Language		
• The elements of active listening • Turn-taking techniques • Initial engagement techniques • Verbal and nonverbal communication		• Use gestures, actions, body language, and words to communicate needs and to express ideas • Interact effectively with peers and adults in familiar social settings • Follow classroom directions and routines • Use context clues • Listen respectfully to others		
Critiquing Language		Critiquing Language		
• Tone variations • Different moods • Different audiences • Names of emotions		• Identify when actions (verbal/nonverbal) impact others • Identify the meaning of messages by recognizing the mood and tone		
Producing Language		Producing Language		
• The elements of a sentence • Grammatical rules • Word order • Clarity of speech		• Realize that word order can change from one language to another • Use their own grammar style as a part of the process of developing grammatical awareness • Recognize the sounds of a language symbolically represent ideas and objects		
Suggested Time Line	Suggested Learning Experiences	Assessments	Differentiation	Resources
Introduction	When we play we....	Prior knowledge		
	Identify prior knowledge.	assessment of what the students already understand	May provide conflict examples	
	Build essential agreements for the classroom (G#1 & 2)			
Beginning stages	Charades:	Photographs:	Model further	
	Read my face or read my body. Can I	Use these in a photo reflection	with students who may be	

	express emotions or ideas without words? (G#4)	with students.	having difficulties or ask for suggestions from students of other ways the game may be played.	
Throughout the unit	Nonverbal cooperative games (G#2 & 4) 1. Mirrors: In partners, each student will observe and mimic the movements of a partner (nonverbal), as in the mirror image. 2. Sit down: A game where the group works together to assure that the whole group is seated on chairs, one at a time. There can be no verbal communication and students must not use gestures or eye contact. In this game, students learn to watch for subtle clues to work cooperatively to win the game.	Observation and group reflection to discuss success of game and the importance of nonverbal communication. Observation and group reflection to discuss success of game and the importance of nonverbal communication.	Pair students with older buddies if possible. Model the game to assist EAL learners to understand directions. It may take students several attempts, even several days, to complete the game. If students really struggle, you can model some strategies.	Timer Chair for each student placed in a circle
Suggested Time Line	Suggested Learning Experiences	Assessments	Differentiation	Resources
	3. Loopy Hoop: Students will pass a Hula-Hoop around the circle while holding hands.	Observation and group reflection to discuss success of game and the importance of nonverbal communication.		Hula-Hoop

Throughout the unit	Conflict resolution and decision making games	Teachers will observe how students work in pairs or larger groups to play the game.	See if students can think of other ways they use to help them solve conflicts
	1) Rock, Scissors, Paper: Students will learn how to play this as a tool for helping them make decisions or solve conflicts (G#2)	Strategy development and whether it is used in the classroom as a tool for decision making or problem solving	
Beginning of unit	Steps to problem solving —Identifying the problem		
	Possible solutions (G#2&3)	What strategies do students already have?	Work further with students, in
	Wheel of choices (follow on activity)	I noticed...	small groups or
	Pick the solution	Students list out what they noticed about tone of voice and body	one to one for those who find it difficult to solve conflicts.
	Do it		
	Listen to <i>The Zax</i> by Dr. Seuss. Ask students to identify the problem. Was there a solution? What could have been a possible solution? Shall we see if our solutions may		<i>The Zax</i> by Dr. Seuss
	have helped the Zax? Reenact the story with the children and use their solutions to solving the problem, taking note of body language and tone of voice.	language during the reenactments	
Beginning of unit	Wheel of choices: A visual for students to use when trying to solve conflicts (G#2 & 3)	Observe whether students use this as a reminder to	Create personal examples for students who may struggle
			Large piece of paper to record their answers of strategies they can use

		help them when solving problems. Do we see the students use the language within their play?	with picking a solution. Personalize by creating their own pictures, or use photographs that represent the different choices on the wheel.	
	Emotional sandwiches: Do we always just jump to anger? What made you become angry? If anger is the bread of our sandwich, what has happened inside for us to be angry? (G#1)	Do students have the vocabulary necessary to describe their anger or how they became angry?	Explore other emotions. Is anger the only emotion that is built up from other emotions?	Cardboard for sandwiches, different ingredients to show how different emotions create our anger
Middle of unit	Reading Story: <i>The Rain Came Down</i> by David Shannon		Reread the story for those children who need to hear it a second time.	Book: <i>The Rain Came Down</i> by David Shannon
Small groups work together to retell the story using picture prompts.	Can students retell the sequence of the story verbally with the use of the picture cards?	Have the story available in the listening center.	Picture prompt cards for each group	
Whole group discussion to identify reasons for conflict and how the	What themes and concepts do the students draw from the story?	Group stronger students with weaker students to support in group work.	Chart paper and markers	

characters in
the story
resolved that
conflict.
(G#2)

Suggested Time Line	Suggested Learning Experiences	Assessments	Differentiation	Resources
Middle of unit	I messages (an activity to learn how to use an I statement to help communicate how they feel); Role-play different examples using puppets. (G#1&3)	Observation as well as reflection after each partner team models its I message.	Partner younger students with an older student. EAL students can have vocabulary given to them with structured support by teacher.	I message chart
Throughout the unit	Present stories in multiple ways, e.g., through story telling, puppets, dramatic play. Take on character roles and expression (words and body language) through the stories. (G#3)	Responses from students connected to the stories	Students develop their own stories or role-play about stories or situations familiar to them.	<i>Hey, Little Ant</i> by Hannah Hoose
Throughout the unit—leading up to the culminating task	Set the stage: Present pretend scenarios and strategies for solving disagreements using problem solving techniques. (G#2&3)	Talking through suggestions for the situation. Identifying the situation: What was the problem? What ways could we go about solving the problem?	Invite students in to take on roles in the situation. What are their suggestions?	https://www.kidsmatter.edu.au/families/about-friendship/resolving-conflict/resolving-conflict-how-children-can-learn-resolve

Throughout the unit	Learning about sentences: 1) What is a sentence? Students use colored blocks to represent each word in a small sentence.	Observation and reflection during the activities	Partner younger students with an older student. EAL students can have vocabulary given to them with structured support by teacher.	Colored wooden blocks. Sentence strips cut into individual words.
	2) Ask the children to clap, knock on the floor, or click their fingers every time they come to the end of the sentence.	Observation while students are playing will give a teacher the opportunity to assess if students can identify the end of the sentence.	Model this game for students who might need a little more support.	
	3) Finish my sentences: With a partner students will start a sentence and have the partner finish the sentence and then clap, knock, or click when they have completed the sentence.	While playing this game, the teacher can surmise instantly if a student understands the difference between a sentence and a statement.	Partner younger students with older students. EAL students can have vocabulary given to them with structured support by teacher. For EAL students teacher can provide picture vocabulary cards or use sentences from a book that was	

just read together.		
4) Is it a question? This is a game that students play with the teacher. The students in small groups have a craft stick with a foam question mark attached. When they identify the sentence as a question, they raise the question mark. After some practice, the students can partner to identify other examples of questions vs. statements.	Because this game is played with the teacher and in a small group, the teacher is able to observe as well as record on paper the sentences that students build together.	Craft sticks with foam question marks glued to them. One per child in a small group lesson.
5) Who, What, Where, When: With a beach ball that has sides marked as who, what, where, and when, the children sit in a circle and roll the ball back and forth until they have the elements of a sentence. If a child rolls it and it lands on a category already written down by the teacher, the ball is passed on to the next child. Throughout the game, students repeat the elements of each sentence until they have completed it. (G#5)	Partner younger students with older students. EAL students can have vocabulary given to them with structured support by teacher.	Inflatable beach ball or big dice on which is written Who, What, Where, and When. Chart paper or table-top whiteboard and marker.

NB: It is important to note that the classroom environment plays a large role in learning within the Early Years. The teachers use the environments to enhance the understanding of the learning experiences. Examples of this would be the role play area: a place where students can engage in scenarios that may reflect the stories they are reading or have heard or situations that have occurred in class.

Teachers in Early Years are always taking individual and classroom observations. These observations, whether written, photographed, or videoed, are used for assessing students’ understandings. These would be

helpful tools to use alongside evidence from the learning experiences.

Culminating Unit Assessment

What (the unit focus): Investigate building language through play

Why (the generalizations): **Our actions, tone of voice, and words express our feelings and ideas**

How (the engaging scenario for students):

You are the audience. While watching and listening to these role plays, think about how the tone of the voices being used is expressing the actor's feelings. What about the actor's actions? How do you think the actors were feeling? What actions, gestures, and cues made you think that?

Now you are going to have the chance to be the actor. In a small group, act out your given story focusing on appropriate and inappropriate words, tones of voice, and actions for the situation.

As a group, reflect on how the different actions, tones of voice, and words changed the feelings and messages being expressed. (Teacher will record this.) Can you make any suggestions for how a different action, tone of voice, or word might have changed the way you felt?

One on one with the teacher, work on a photo reflection, explaining your understanding of how actions, tone of voice, and our words express our feelings and ideas. Why should we be aware of the tone of our voice?

Scoring Guide

Primary Grades

4 Meets all standard criteria plus:

Examples:

Gave detailed descriptions of differing tones of voice and the emotions they convey.

Provided clear and detailed identification of feelings and tones of voice

✓ If met

S T A N D A R D	<u>Understanding</u>	
	Thorough explanation of how actions, tones of voice, and words express our feelings and ideas	
	<u>Content</u>	
	Able to identify feelings expressed through examples of actions, e.g., gestures and cues	
	Demonstrated examples of different tones of voice in different situations	
	Explained how tones of voice change feelings and messages	
	Connected understandings through role play and through photo reflection	
	<u>Process</u>	
	Clear articulation	
	Identified and made use of appropriate intonation, tone, and mood	
	Used appropriate word choice	

2 Meets 5/8 of the criteria in the standard

1 Not able to score; does not yet approach standard

Lesson Resources

Cooperative Games

1. Mirrors

Pair up students. Explain that they will play two rounds. In each round, one person will be the leader and one person will be the follower. Have pairs choose the leader for the first round. Explain that the leader will have 60 seconds, during which that person should move slowly and deliberately and during which the follower must mimic the leader's movements, as if the follower is a mirror image. Leaders may make faces, wave their arms or legs, turn side to side, stand on one foot, retie a shoe, or anything they can imagine (within reason). It should be a silent activity, and leaders should move slowly and smoothly enough so that their "mirror images" can follow along in sync. After 60 seconds, have players switch roles.

Variation

After each person has had a chance to be the leader and the follower, give groups a minute during which neither person is designated the leader.

2. Loopy Hoop

Have the children stand in a circle, holding hands. Ask two children to drop hands for a moment and then place a Hula-Hoop or loop of rope between them. When they rejoin hands, the hoop will be hooked over their arms, dangling between them. Tell the children that, from this point on, none of the children can drop their hands. Explain that the goal of the game is to pass the hoop around the circle. Each child maneuvers the loop over her own head and steps through it, possibly with help from others in the circle. After the children pass the hoop successfully around the circle in this way, play can continue in the other direction.

Conflict Resolution or Decision Making Games

1. Rock, Scissors, Paper

The players usually count aloud to three, each time either raising one hand in a fist or swinging it down on the count. On the fourth count, the players change their hands into one of three gestures—rock, scissors or paper—which they then "throw" by extending it towards

their opponent.

Activities

I Messages:

This activity helps children to learn the skill of recognizing emotion, in themselves and in others. This can be done using I messages. Kids can start with an I statement, such as “I feel sad when you don’t play with me.” This helps them identify their emotions, and they would stop blaming others. This will also guide them through talking out their conflict, before starting or discussing possible solutions.

Story—*The Rain Came Down* by David Shannon:

This story helps students learn to be aware of their surroundings and the effects they can have on a person’s feelings and reactions.

Vocabulary: Words that may need to be discussed before reading the story include *squawk*, *yowled*, *squirmed*, *beauty parlor*, *barbershop*, *bickering*, and *ruckus*.

Introduction: Questions to ask before reading include the following.

- “By looking at the cover, what can you tell me about how the characters are feeling in the story?”
- “What is causing them to react this way?”
- “How do you feel when it rains?”
- “How do you feel when it is sunny?”

Questions During Reading:

- “What is an action word?”
- “What is causing the entire ruckus in the story?”
- “How are things changing at the end of the story?”
- “What might happen if the rain gets harder or stops coming down?”

Unit Example: English Language Arts

Grade Level: Grade 7 Unit Number: Three

Title: See With Their Eyes, Feel With Their Heart

Conceptual Lens: Perspective

Unit Length 9 Weeks

Author(s): Tiffanee Brown, Grade 7-8 Language Arts, Bay View School, Burlington-Edison School District, Washington

Unit Overview:

*“You never really know a man until you understand things from his point of view, until you climb into his skin and walk around in it.” **Harper Lee, To Kill a Mockingbird***

Have you ever thought about what the world would look like through someone else’s eyes? Or what your life would be like if you were born in a different city, or even country? Would you be a different person given a different set of life circumstances?

In this unit, we will explore the relationship between point of view and compassion through reading and discussing stories, poems, personal essays, and informational texts. We will conduct mini-inquiries to dig deep into the author’s craft and to examine how authors use writing to communicate ideas to the reader. This will also help us see how considering the point of view of others impacts our beliefs, actions, and understanding of important issues. In writing teams, you will create a meaningful story that teaches your reader about a realworld issue told through multiple points of view. You will use all of the things you’ve discovered about effective writing to help your reader “climb inside” and see the world through the eyes of your character. Our exploration will conclude with a publishing celebration in which we read the books created by the writing teams.

Technology Integration:

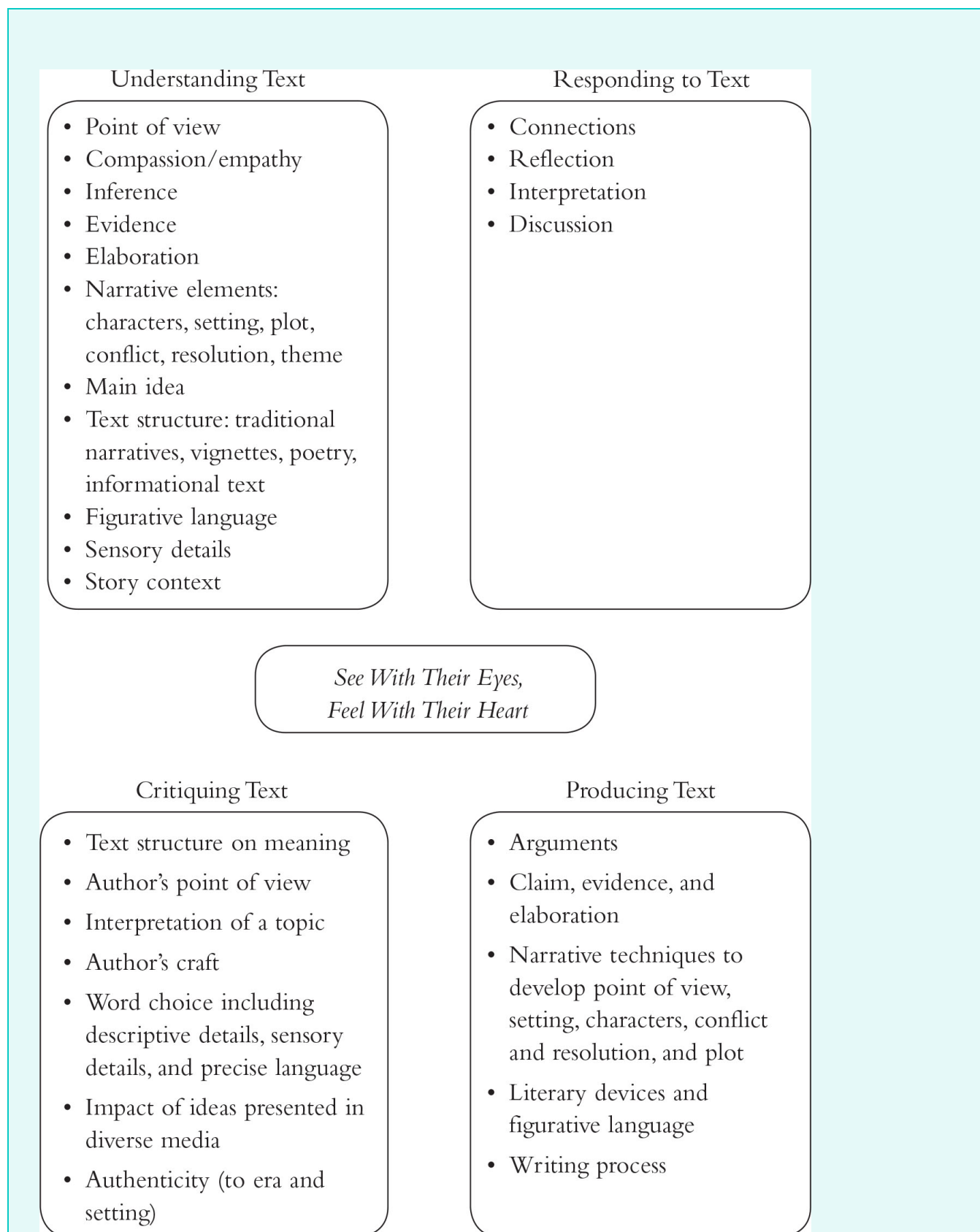
- Word Processing
- Digital Collaboration through Google Docs

Standards or national curricula addressed in this unit:

This unit has been cross checked and aligned with the Common Core State Standards.

Unit Title: See With Their Eyes, Feel With Their Heart

Conceptual Lens: Perspective Grade Level: 7



Grade Level:7 Title:See With Their Eyes, Feel With Their Heart

Generalizations	Guiding Questions: F = Factual; C = Conceptual; D = Debatable
1. Considering the point of view of others builds compassion and understanding (UT, RT, PT).	1a. What is point of view? (F) 1b. How did _____ (character) feel about _____ (character) before getting to know him/her at a personal level? (F) 1c. What is the relationship between point of view and compassion? (C) 1d. Why do we need to consider the perspective of others? (C) 1e. What would the world be like if we never took the time to consider the perspective of others? (D)
2. Different perspectives lead to multiple representations and interpretations of a similar topic, idea, or event (UT, RT, CT).	2a. Why do people interpret the same topic or idea in different ways? (C) 2b. When reading a story, how do you know how a character interprets and feels about an event? (C) 2c. How might _____ (character) interpret one event in the story differently from _____ (character), and how do you know? What role do their points of view play in this interpretation? (F) 2d. How does viewing an experience through the eyes of different people help you understand an issue?(C) 2e. What makes a perspective credible? 2f. Can one perspective ever be considered better than another? Explain (D)
3. The setting of a story shapes the character's perspective and actions (UT).	3a. How do authors develop the setting of a story? (F) 3b. What was the time frame of the story _____? (F) 3c. How does the setting (including location and era) of a story impact a character's thinking, beliefs, and actions? (C) 3d. How can background knowledge of a story's setting and time frame affect a reader's interpretation of a story? (C) 3e. Does our "where" influence who we become? How does our environment impact our thinking, beliefs, and actions? (C)

	3f. Would you be a different person if you grew up in a different location? (D)
4. Authors use narrative techniques, such as dialogue, pacing, and description, to develop experiences, events, and/or characters to build a complex story (UT, PT).	4a. How is a story told through 13 perspectives different than a traditional story? (F) 4b. How do authors represent a progression of time in stories? (F) 4c. How do authors give characters unique voices? (F) 4d. How does the author develop point of view within the story or poem? What evidence leads you to this conclusion? 4e. How do elements of a story interact? If one element is altered, what happens to the story? (C)
5. Precise words and sensory language capture action and convey experiences to the reader (UT, RT, PT).	5a. What is precise language? (F) 5b. What are sensory details? (F) 5c. Why does word choice matter? (C) 5d. How do authors communicate feelings, ideas, and details of life experiences using very few words? What details create the biggest impact? (C) 5e. How do precise words and sensory details draw in and engage the reader? (C)
6. Story themes reveal messages about life that an author wants the reader to think about in order to broaden perspective (UT, PT).	6a. What is theme? (F) 6b. How do authors develop the theme of a text? (C) 6c. How did this author develop the theme of this story or poem? (F) 6d. What strategies do readers use to understand the theme of a text? (F) 6e. What makes a reader connect with the theme of a text? (C) 6f. What makes a story enduring? (D)
7. Supporting inferences with textual evidence strengthens the credibility of the analysis (UT, CT, PT)	7a. Why do I need evidence when sharing my interpretation of a story or poem? (C) 7b. How do I make my evidence more effective, e.g., sound good or flow? (F) 7c. What are some strategies to deepen the analysis in my writing? (F) 7d. What happens if readers don't use evidence to support a story analysis?

(C)

Note: Generalizations are coded to different web strands (UT, RT, CT, Pt) to show they collectively represent all dimensions of a process.

Critical Content and Key Skills

Critical Content by Strands

Students will be able to...	Key Skills
Students will know... Understanding Text - Point of view and the role it has in a text - How point of view influences voice - Text structure: traditional narrative, vignette, informational, poetry - Narrative elements: characters, setting, plot, conflict, resolution, theme, and how the elements interact to develop a story - Meaning of compassion - Importance of supporting inferences with evidence - Meaning of theme, and how authors develop theme - Role of figurative language and sensory details in building meaning Responding to Text - Readers make connections with	Understanding Text: <u>CCSS.ELA-LITERACY.RL.7.1</u> Cite several pieces of textual evidence to support analysis of what the text says explicitly as well as inferences drawn from the text. <u>CCSS.ELA-LITERACY.RL.7.2</u> Determine a theme or central idea of a text and analyze its development over the course of the text; provide an objective summary of the text. <u>CCSS.ELA-LITERACY.RL.7.3</u> Analyze how particular elements of a story or drama interact (e.g., how setting shapes the characters or plot). <u>CCSS.ELA-LITERACY.RL.7.4</u> Determine the meaning of words and phrases as they are used in a text, including figurative and connotative meanings; analyze the impact of rhymes and other repetitions of sounds (e.g., alliteration) on a specific verse or stanza of a poem or section of a story. Responding to Text: <u>CCSS.ELA-LITERACY.SL.7.1</u> Engage effectively in a range of collaborative discussions (one-on-one, in groups, and teacher-led) with diverse partners on Grade 7 topics, texts, and issues, building on others' ideas and expressing their own clearly.

texts	<u>CCSS.ELA-LITERACY.RI.7.3</u>
• Strategies for reflection • Strategies for collaborative discussion	Analyze the interactions between individuals, events, and ideas in a text (e.g., how ideas influence individuals or events, or how individuals influence ideas or events).
Critiquing Text	Critiquing Text
	<u>CCSS.ELA-LITERACY.RL.7.5</u>
• Impact of text structure on meaning • Author's point of view and interpretation of a topic • Meaning of author's craft • Impact of intentional word choice including descriptive details, sensory details, and precise language • Impact of ideas presented in diverse media	Analyze how a drama's or poem's form or structure (e.g., soliloquy, sonnet) contributes to its meaning
	<u>CCSS.ELA-LITERACY.RL.7.6</u>
	Analyze how an author develops and contrasts the points of view of different characters or narrators in a text.
	<u>CCSS.ELA-LITERACY.RI.7.4</u>
	Determine the meaning of words and phrases as they are used in a text, including figurative, connotative, and technical meanings; analyze the impact of a specific word choice on meaning and tone.
Producing Text	Producing Text
	<u>CCSS.ELA-LITERACY.W.7.1</u>
• Role of claim, evidence, and elaboration in supporting arguments • Narrative techniques to develop point of view, setting, characters, conflict and resolution, and plot • Role of literary devices and figurative language in developing stories • Stages of the writing process	Write arguments to support claims with clear reasons and relevant evidence.
	<u>CCSS.ELA-LITERACY.W.7.3a-e</u>
	Write narratives to develop real or imagined experiences or events using effective technique, relevant descriptive details, and well-structured event sequences.
	<u>CCSS.ELA-LITERACY.W.7.5</u>
	With some guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on how well purpose and audience have been addressed.

Grade Level: 7 Unit Title: See With Their Eyes, Feel With Their Heart

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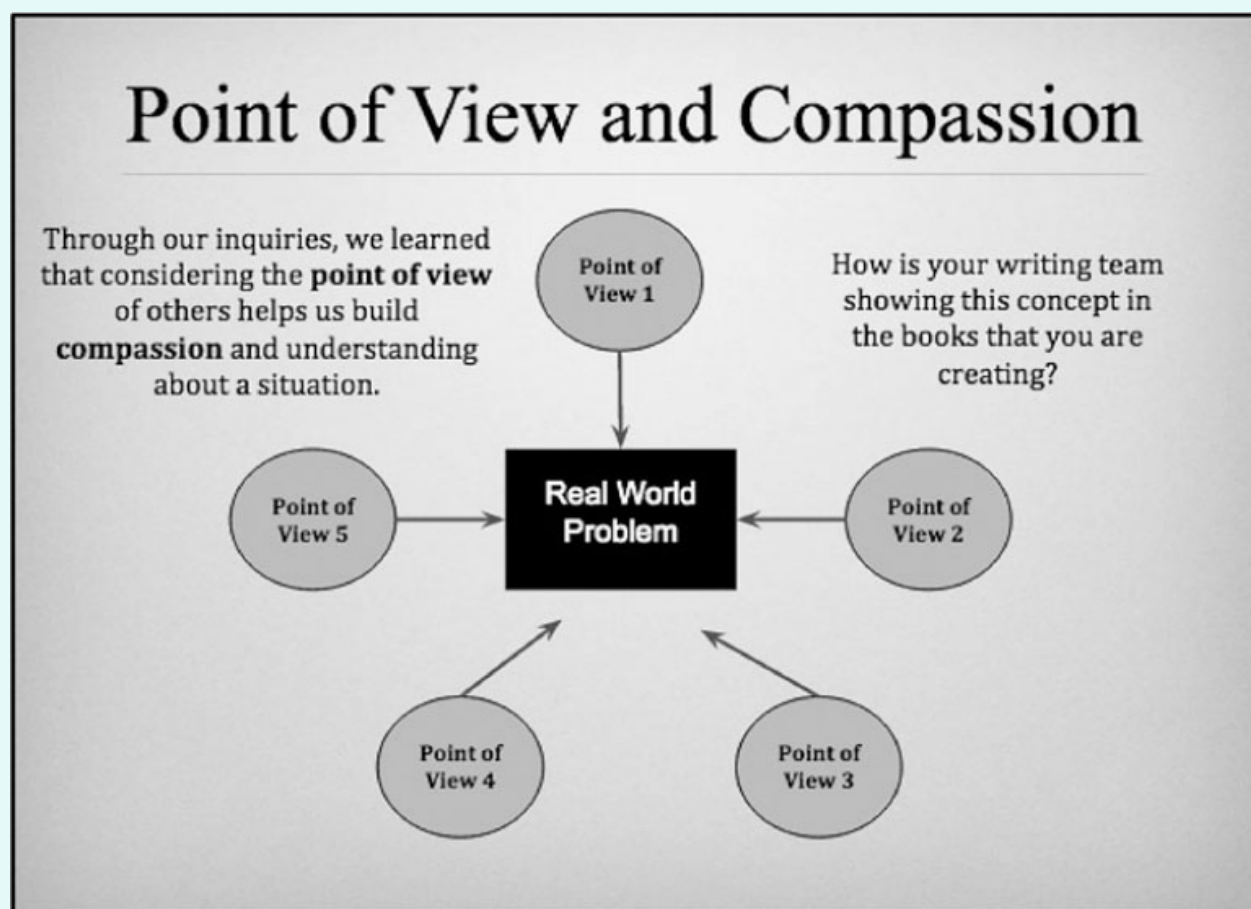
Culminating Unit Assessment

Grade Level: 7

Unit Title: See With Their Eyes, Feel With Their Heart

What (the unit focus): Students will explore the relationship between point of view and compassion...

Why (the generalization): in order to understand that considering the perspective of others builds compassion and understanding.



How (the engaging scenario for students):

Your challenge is to identify a realistic issue or problem and create a story that shows the characters working through that situation. Each chapter of your story will be told through a different point of view so that the reader is able to understand the issue deeply. Find your inspiration by thinking about real-life situations and problems about which you feel strongly.

Upon completion of your story, you will reflect individually on your learning from this unit and from the work you did yourself and with your team. How do you think that presenting the perspectives of each character in your story will help build compassion and understanding in readers about the story's problem?

Steps:

- Identify a situation, problem, or issue that may be encountered by a young person.
- Identify where the problem occurs or takes place.
- Create your cast of characters to cover the different perspectives and points of view that are involved with the issue. Each person on your writing team will write from the point of view of one character. You can have primary and supporting characters, as in *Seedfolks*.
- Plan the story. How can you bring your characters together to solve the problem or overcome the issue?
- Write the story.
- Meet as a writing team.
- Revise and edit the story.
- Create artwork.
- Publish your books.
- Individual author's reflection: Write 1-2 paragraphs discussing how you believe your story will help readers consider the perspective of others and build compassion and understanding about your

story’s problem or issue. Give a real-life example of the consequences that can result when the perspective of others is not considered.

- Celebrate!

Culminating Unit Assessment Rubric

Exceeding	Meeting Standard (3)	Not yet,
Standard	Circle descriptor if met	but I will!
(4)		(2)

Process	Narrative Focus
	The author • engages and orients the reader by establishing a context and point of view and introducing a narrator and/or characters; • demonstrates an event sequence that unfolds naturally and logically; • uses narrative techniques, such as dialogue, pacing, and description, to develop experiences, events, and/or characters; • provides a conclusion that follows from and reflects on the narrated experiences or events.
	Language and Vocabulary
	The author • uses precise words and phrases, relevant descriptive details, and sensory language to capture the action and convey experiences and events; • uses a variety of transition words, phrases, and clauses to convey sequence and signal shifts from one time frame or setting to another.
	Conventions

The author

- demonstrates command of the conventions of standard English grammar, capitalization, punctuation, and spelling.

Collaboration

Our group consistently worked together to accomplish our goals by

- working through our challenges in respectful ways;
- compromising;
- honoring the ideas and contributions of all group members;
- offering thoughtful and constructive feedback on our writing.

Theme

Our story effectively develops a theme by

Content

- addressing a realistic problem or issue that a seventh grader may encounter;
- allowing the reader to view the issue through the eyes of a variety of perspectives;
- teaching the reader an important life lesson.

Understanding of Generalization

My individual author's reflection

Understanding

- thoroughly explains how presenting the perspectives of each character in my story helps to build compassion and understanding in readers about the story's problem
- references and reflects on concepts explored during the unit
- discusses the consequences that can occur when the perspective of others is not considered

Score: ____/41 - Not enough work provided to score

Concept-Based Lesson Planner Template

Unit Title _____ Instructor _____

Subject _____ Grade Level _____

Lesson Number _____ Lesson Time Frame _____

Lesson Opening (communicated to students at lesson onset):

Learning Targets: What Students Will *Understand* (Generalizations), *Know*, and Be Able to *Do* (Skills)

Generalization(s) (Students will understand that...)

Note: Generalizations may apply to one or a series of lessons.

1.

Guiding Questions

– 1a.

– 1b.

– 1c.

Strand Critical Content (Know) Key Skills (Able to Do)

Learning Experiences

Differentiations

Materials/Resources

Assessments

Closing

Teacher Notes

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Concept-Based Lesson Plan Example

Chemical Bonding Lesson (Grade 11)

Unit Title: Chemical Bonding and **Instructor:** Mona Seervai

Structure: All About the Shape!

Subject: Chemistry **Grade Level:** 11

Lesson Number: 4 **Lesson Time Frame:** 3 class periods

Lesson Opening (communicated to students at lesson onset):

Molecules occupy a three-dimensional world! We may forget this after seeing so many depictions of molecular structures on a two-dimensional page. The Lewis electron-dot structures you have learned to draw have no geometrical significance other than depicting the order in which the various atoms are connected to one another.

In this lesson, we will be using physical models as well as computer simulations to create the 3D models. Understanding the concept of the electron domain and using the VSEPR model can predict the geometry of the bonds around a given atom. At the end of the lesson, you will be able to calculate the bond angle and predict the shape of molecules.

Learning Targets: What Students Will Understand (Generalizations), Know (Factual Content), and Be Able to Do (Skills)

Generalization(s)

Students will understand that...

1. **Sharing electrons causes covalent compounds to form.**

Guiding Questions

- 1a. What is an electrostatic force of attraction? (F)
- 1b. Which electrons are also designated as valence electrons? (F)
- 1c. How do Lewis diagrams show the arrangement of electrons? (F)
- 1d. How are the different types of bonds (single, double, triple) created? (C)

2. Electron domains in the valence shell may predict molecular shape.

Guiding Questions

- 2a. What is an electron domain? (F)
- 2b. Why do atoms have a tendency to gain electrons in the valence shell? (C)
- 2c. How does repulsion of electron pairs determine the shapes of molecules? (C)
- 2d. How does the creation of 3D structures give an understanding of the bond angles? (C)
- 2e. Does the creation of 3D structures change the significance of Lewis diagrams? Why or why not? (D)
- 2f. How do scientists use criteria to assess the validity of a scientific theory? (D)
- 2g. To what extent does the language used in scientific terminology deviate from common usage? (D)

Strand	Critical Content (Know)	Key Skills (Able to Do)
Covalent Compounds	• Electrostatic force of attraction between shared electrons and nucleus • Types of bonds	• Construct electron-dot structures • Predict bond angles from molecular geometry
Molecular Structure	• Definitions of Lewis structure and valence electron • Octet rule and reasons for stability	• Use structure to deduce properties of compounds • Predict shapes of species depending on electron pairs

Learning Experiences	Differentiations
Review covalent molecules: Watch a video from the Royal Society of Chemistry on Lewis structures and bonding. Identify the formation of single, double, and triple bonds.	Discussion on the video supplemented by material from textbook.
Practice drawing Lewis structures of hydrogen, oxygen, chlorine, nitrogen, water, and methane.	Worksheets with different levels of scaffolding.
An inquiry process is initiated with the use of molymod organic molecule sets. Students work in pairs to create the following 3D molecules: oxygen, ammonia, and water.	Mixed ability pairs.
Draw the molecules methane, ammonia, sulphur dioxide, and carbon dioxide using ChemSketch.	Vary the instructions. Individual work on "ChemSketch" with peer and teacher guidance.
Copy the drawing to the 3D viewer. For each one, measure the bond angles, draw the shape on the worksheet, and decide the geometry of the molecule. Complete the worksheet.	

Materials/Resources

Video from:

<http://www.rsc.org/learn-chemistry/resource/res00001370/chemistry-vignettes-bonding-theory-and-vsepr#!cmpid=CMF00003191>

Chemistry for the IB Diploma Programme (2nd edition), Pearson Baccalaureate, by Catrin Brown and Mike Ford (2014).

ChemSketch (version 11.0) – free software to be downloaded:

(http://www.acdlabs.com/products/draw_nom/draw/chemsketch/)

Worksheets with Lewis structures of covalent molecules

Web information: <http://www.chem1.com/acad/webtext/chembond/cb05.html>

Assessments

1. Formative assessment: Observe students' accuracy in drawing Lewis structures. Check the worksheets.
2. Initiate a class discussion on making 3D models from molymod set.
3. Use ChemSketch to draw molecules and look at molecules in the 3D viewer.
4. Assign this performance task at the end of the lesson:
You are a scientist making a new judgment about a structure.

Investigate the molecule "Ozone."

What you will do:

- a. Draw the Lewis structure.
- b. Estimate the bond angle – use the VSEPR model as well as "ChemSketch."
- c. Use the information to predict the shape of the molecule.

You are to write a description giving reasons for your judgment, draw a model, and create a 3D model of your molecule.

Extension: Add one more polyatomic ion like CO_3^{2-} and NO_3^- or HCO_2^- (these have a resonance structure, and students can inquire into a new concept).

Closing

The VSEPR model is powerful in its simplicity and ability to explain molecular shape. However, as with all scientific models, it is limited because it relies on electron repulsion only and not on the orbitals and bonds themselves.

Teacher Notes

This lesson is part of a unit on chemical bonding and structure. The inquiry process using molymod (or

equivalent) models generates a high level of interest. The use of computer software to predict shapes of molecules also sustains interest. The discussion on validity of scientific models can be a link to the “Theory of Knowledge” in the International Baccalaureate Diploma Programme.

Source: *Mona Seervai, Mumbai, India*

Learning Experiences: Dr. Anuradha Sridha, Grade 11 Chemistry Teacher, Bombay International School © 2012 Lois Lanning- Lesson Planner Template

Checklist for Evaluating Concept-Based Curriculum Units

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Unit Web and Unit Overview:

_____ 1. Is the unit title clear so that anyone reading it would know the topic of study?

Strong title: The Impact of Technology on Community Life

Weak title: Widgets and Whirligigs

_____ 2. Is the unit title neither too narrow nor too broad?

Too broad: Life; Patterns; Systems

Too narrow: The Earth's Crust; Manifest Destiny; Two-Digit Multiplication

_____ 3. Is there a suitable conceptual lens attached to this unit of study (either built into the title or shown at the top of the unit web page)?

- Does the lens provide a clear focus for the unit title?
- Does the lens relate to, and reflect, a deeper understanding of the unit under study?
- Does the lens suggest transferable connections to other subject areas or topics?

_____ 4. Do the strands represent the major areas of study for this unit?

_____ 5. Are the critical subtopics, as well as subconcepts, listed under each strand (especially in heavy content subjects such as history)?

_____ 6. Do the subconcepts and subtopics under each strand represent the requirements of academic standards or national curricula and provide a complete treatment of the strand for this grade level?

_____ 7. Does the unit overview provide a concise and engaging summary of the unit of study?

_____ 8. In process-driven subjects, do the strands show the concepts derived from the unit processes, strategies, and/or skills?

Generalizations (UNDERSTAND):

_____ 1. Are the generalizations developed from the required standards or national curricula, and do the generalizations use the most important concepts for the particular unit of study?

_____ 2. Do the generalizations reflect the most important conceptual understandings that students are to take away from the study?

_____ 3. Do generalizations enable students to transfer knowledge to a broader context?

_____ 4. Are the generalizations clear—can fellow teachers immediately relate the generalization to the unit content?

_____ 5. Are the generalizations devoid of Level 1 verbs (no-no verbs): *affect*, *impact*, *influence*, *is*, *are*, and *have*.

_____ 6. Do the unit generalizations collectively represent all the strands of the unit web?

_____ 7. In content/concept-driven discipline units (e.g., social studies, science, mathematics), are there any essential process generalizations that need to be included?

_____ 8. Do the generalizations use present tense, active verbs?

_____ 9. Do the verbs avoid the passive voice (e.g., “can be caused by”). Note: If a verb is in the passive voice, flip the sentence, putting the object of the sentence (the nouns after the “by”) at the beginning, as the sentence’s subject, to create a present tense, active verb. For example, “Feelings of anxiety, confusion, and anger can be caused by severe disruption of a community’s social and economic infrastructure” becomes “Severe disruption of a society’s social and economic infrastructure can cause feelings of anxiety, confusion, and anger.

_____ 10. Do the generalizations exclude the use of pronouns, proper nouns, or topics that would lock the ideas in time, place, or situation?

_____ 11. Do the generalizations include a qualifier (often, can, may) when a statement does not hold true in all cases?

____ 12. Do the generalizations avoid making a “value statement” (signaled by words such as *should* or *must*)?

____ 13. Does each generalization represent a statement of relationship between at least two concepts joined by a strong verb?

Guiding Questions:

____ 1. Are there mixes of three to five factual and conceptual questions for each generalization? Note: Level 1 verbs are fine in questions—but not in the generalizations.

____ 2. Are there a few debate questions for the entire unit? Note: Debate questions can be open ended, or they may include proper nouns, past tense verbs, and Level 1 verbs. These questions must be debatable.

____ 3. Do the questions guide students thinking from facts and skills to the target generalization (generally inductive teaching)? Note: It is fine to pick up exact language from the generalization and build it into the conceptual question. Tip: It is helpful to have teachers code each question (F, C, D).

____ 4. Do the conceptual and debatable questions invite student inquiry?

Critical Content (KNOW):

____ 1. Is the critical content a list of essential factual *knowledge* rather than a set of content objectives with verbs? (*Know* is the verb for the entire list.)

____ 2. Is the critical content written as a list of essential factual *knowledge* rather than as generalizations?

____ 3. Does the critical content provide support for the generalization and reflect other important *knowledge* relative to the topic under study?

____ 4. Does the critical content that is not tied directly to generalizations still reflect important *knowledge* for the unit topic?

Key Skills (DO):

- _____ 1. Are the key skills drawn from standards documents or national curricula?
- _____ 2. Do the skills represent a range of thinking processes?
- _____ 3. Are the skills written so they can be transferred across many different applications?
- _____ 4. Do the skills avoid naming a specific topic? Note: A skill can be written so that it transfers within a discipline (e.g., “Manipulate polygons to illustrate transformations”), but it cannot name a particular topic (e.g., “Evaluate primary and secondary source documents to learn about Native American culture”).

Assessment:

- _____ 1. Does the culminating assessment address the three critical components of a Concept-Based unit: *understand* (generalization), *know* (critical factual content), and *do* (skills)?
- _____ 2. Does the assessment pick up *language* from the target generalization that reflects deep, conceptual understanding?
- _____ 3. Are the “what, why, and how” components correct? (What = unit title; Why = most important unit generalization; How = the performance) Note: Encourage participants to write an authentic (real world) performance through an “engaging scenario.” RAFTS, or some other tool, may be used to assign a role, audience, format, topic, and strong verbs and adjectives.
- _____ 5. Is the assessment task rigorous, and does it require students to apply—not just regurgitate—learning to a new context, situation, or problem?
- _____ 6. Are there a variety of assessment types throughout the unit to provide evidence of what students *understand*, *know*, and can *do*?

Scoring Guide:

- _____ 1. Does the scoring guide clearly describe the criteria for what students are expected to *understand, know*, and *do* (skills)?
- _____ 2. Does the scoring guide weight the different criteria?
- _____ 3. Does the scoring guide encourage student self-assessment?
-

Learning Experiences:

- _____ 1. Do the learning experiences allow students to explore the generalizations, knowledge, and skills in a way that is authentic to the field or discipline?
- _____ 2. Do some of the learning experiences prepare the students for the culminating task(s)?
- _____ 3. Do some learning experiences address the other generalizations, knowledge, and skills not addressed in the performance task?
- _____ 4. Do the learning experiences contain enough description so that another teacher could easily follow the plan?
- _____ 5. Are the learning experiences organized for maximum engagement and effectiveness?
- _____ 6. Are the learning experiences meaningful and worth the time?
- _____ 7. Are there too many learning experiences that address the same topic?
- _____ 8. Do some of the learning experiences require students to utilize and develop inquiry skills?
- _____ 9. Do student performances meaningfully integrate various curricula or areas of study?
- _____ 10. Are some of the learning experiences differentiated to student learning needs and interests?
- _____ 11. Tip: It is helpful to code learning experiences with Generalizations that they specifically teach toward (e.g., G#1, G#2, and so on).

Mathematics Generalizations for Secondary Grade Levels

Jennifer Wathall's 2016 book **Concept-Based Mathematics** provides many excellent generalizations to help teachers in Grades 6–12:

Trigonometry

- For right-angled triangles, the area of the square drawn from the hypotenuse represents the sum of the areas of the squares drawn from the other sides.
- Similar right-angled triangles share a common acute angle.
- Similar right-angled triangles share common ratios of corresponding sides.

Functions

- Variables connected in direct proportionality imply a linear relationship.
- A function represents a mapping (rule) that assigns each input (domain) with one output (range).
- Linear functions show relationships that exhibit a constant rate of change.
- Transformation and translation of a linear function changes the rate of change and the initial value.

Circle Geometry

- The ratio of circumference to diameter in all circles represents a fixed constant, π .
- Summing the limit of sectors forms a rectangle, which formulates areas for circles.
- Angles inscribed in semicircles correspond to right angles.
- The radian measure of angles represents constant proportionality between the radius and arc length.

Calculus

- The derivative can be exemplified physically as a rate of change and geometrically as the gradient or slope function.

- Tangents identify the instantaneous velocity or acceleration at a particular point in time.
- The equation of the tangent and the normal demonstrates the direction of motion and forces respectively.
- Differentiation and integration uncover inverse processes of each other.
- Theorems in calculus allow the evaluation of a definite integral without having to go back to the definition and take the limiting sum of a bounded area.
- Rotating a bounded area around an axis produces a solid whose volume can be determined using integral calculus.

Miscellaneous Topics:

- Logarithm laws give a means of changing multiplicative processes into additive processes, and this can provide the means to find inverses of exponential functions, which represent continuous compounded growth.
- Statistical analysis and graphic displays often reveal patterns in seemingly random data or populations, enabling predictions.
- The scalar product between vectors represents the angle between the vectors and magnitude of the vectors.

Process Generalizations

- Mathematicians create different representations—table of values, algebraic, geometrical—to compare and analyze equivalent functions.
- The revision of a mathematical model or substitution of data may enhance or distort an accurate interpretation of a problem.

Source: Wathall, Jennifer. (2016). *Concept-Based Curriculum and Instruction in Mathematics: Teaching for Deep Understanding in Secondary Schools*. Thousand Oaks, CA: Corwin.

Early American Colonization: History Unit Web

Interactions/Culture (conceptual lens)

History/Culture

- European *explorations*, 15th to 17th centuries—factors and features
- *Cultures and interactions of populations (beliefs and values; perceptions)*
- *Indigenous cultures in the Americas*
- The differences in *colonization* by the *mother countries*
- *Interactions with other colonies and the native Americans (beliefs and values; domination, Characteristics of the 13 colonies)*
- The *role of religion* in the English colonies including *governance*, relation to the Americas by England, France, and Spain, including Puritanism and the Great Awakening

Early American Colonization: Beginnings to 1763

Government

- *Patterns of political organization* after 1450
- *Governance – the role of religion*
- *Conflict; perceived oppression by a mother country*

Geography

- *Location of the 13 colonies*
- *Exploration routes*
- *Available resources for the colonies*

Economics

- *Bartering/trade*
- *Economic reasons for exploration*
- *Meeting needs for survival – goods/services*
- *Tariffs/taxation*
- *Competition*

Frequently Asked Questions

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1. *What is Concept-Based Curriculum and Instruction (CBCI)?*

CBCI is a three-dimensional curriculum and instruction design model that frames the facts and skills of subject areas with disciplinary concepts and generalizations. CBCI differs from the traditional two-dimensional model of topics, facts, and lower level skills by raising the bar to include conceptual understandings as essential learning targets.

2. *What is the difference between two-dimensional and three-dimensional curriculum models?*

- Two-dimensional models focus on facts and skills;
- Three-dimensional models focus on concepts, facts, and skills to express deeper conceptual understandings related to the unit of study.

3. *Why should we use a CBCI curriculum design model?*

A CBCI curriculum

- Develops structures in the brain (brain schemata) for sorting, organizing, and patterning incoming information;
- Engages the intellect on two levels—the lower level (factual/skill) and higher level (conceptual);
- Creates “synergistic thinking,” which is the interplay between the lower (factual/skill) level and the higher, conceptual level of thinking;
- Increases motivation for learning by inviting students to think about the facts and skills through a personally engaging “conceptual lens”: for example, considering the current Middle Eastern conflicts through the conceptual lens of *perspectives* or considering poetry through the lens of *text structure* or *imagery*;
- Increases students’ fluency with the conceptual language of a discipline thus promoting richer academic discourse;
- Allows students to transfer their understanding and to see the patterns and connections at the conceptual level; and
- Expands intellectual sophistication by requiring thinking teachers and thinking students.

4. *How does a CBCI curriculum design model change classroom instruction?*

- In a CBCI instructional model, teachers use the facts and lower level skills as *tools* to help students reach deeper understanding of the transferable concepts and generalizations. Facts and skills provide the foundation and support for deeper,

conceptual understanding.

- Instruction focuses on the inquiry process as a means to guide students' thinking deliberately to the construction of a target generalization (conceptual understanding).
- Assessments (formative and summative) not only measure students' progress with facts and skills but also measure students' depth of understanding of the target generalizations.

5. *What are the benefits of CBCI instruction?*

- Students retain the factual knowledge longer because they have processed the information at a deeper level in the brain as they relate lower level facts and skills to concepts and conceptual understandings.
- Students understand processes, strategies, and skills at a deeper level by experiencing the application of the conceptual relationships across multiple contexts. Process-driven disciplines, such as English language arts, world languages, the visual and performing arts, and music, move from simply "doing" skills to understanding the "why" and "how" behind practices.
- Students transfer understandings through the developing conceptual structures in their brain schemata.
- Teachers can compact the burgeoning curriculum around the critical concepts, principles, and generalizations in each subject area.
- Students are personally intellectually engaged and demonstrate greater motivation for learning.

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